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AutoBEM: A scalable framework for nationwide building energy simulation and retrofit evaluation in the United States

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ABSTRACT

This paper presents *AutoBEM*, an integrated, automated framework for nationwide building energy modeling and retrofit evaluation in the United States. Unlike prior UBEM platforms that either rely primarily on representative stock sampling or operate at city scale, *AutoBEM* automates the generation of building-resolved, physics-based EnergyPlus/OpenStudio simulation models at national scale using GIS-derived geometry, prototype-based assumptions, and standardized scalable workflows. Leveraging the Model America dataset and high-performance computing, *AutoBEM* generates and simulates energy models for 122.9 million buildings, representing 97.8% of the U.S. building stock. These models are being made publicly and freely available as the Model America v1.0 (MAv1) dataset. *AutoBEM* supports detailed, building-level assessments of energy consumption, CO_2 emissions, and post-processed anthropogenic heat emissions (AHE), and evaluates 151 energy conservation measures (ECMs) using localized utility pricing and building characteristics. In addition, *AutoBEM* incorporates both typical and future climate conditions through integration with Typical Meteorological Year (TMY) and Future TMY (fTMY) weather data derived from IPCC scenarios.

In a case study of Phoenix, Arizona, *AutoBEM* identified several high-efficiency HVAC upgrades and selected envelope measures with short modeled payback periods (1.5 years) for certain building types and standards. Simulations under future climate scenarios (SSP5–RCP8.5) project an 11.3% increase in electricity use and a 32% reduction in natural gas demand by 2100, underscoring the need for climate-adaptive retrofit planning.

By enabling reproducible, bottom-up, and location-specific analysis at scale, *AutoBEM* provides a step toward a national digital twin of the built environment and supports data-driven screening and planning for decarbonization, resilience, and energy equity.

1. Introduction

Buildings are the largest energy-consuming sector in the United States, accounting for approximately 40% of total primary energy use and about 76% of electricity consumption (U.S. Department of Energy, 2015). Moreover, the residential and commercial sectors collectively generated roughly 31% of total U.S. greenhouse gas emissions in 2022, when considering both direct and indirect sources (U.S. Environmental Protection Agency, 2025). Improving building energy efficiency is therefore essential. The U.S. Department of Energy's Building Technologies Office (BTO) has established ambitious targets, including a 30% reduction in building energy use intensity (EUI) by 2030 and a long-term goal of halving EUI relative to a 2010 baseline (U.S. Department of Energy, 2016). Achieving these objectives economically and effectively mandates robust, scalable modeling platforms capable of capturing regional climate variation, building typologies, and energy pricing.

To date, such high-resolution bottom-up tools remain scarce. Developing the capacity to simulate building performance and evaluate retrofit strategies across the entire U.S. building stock would provide critical support for data-driven policy, investment, and implementation decisions.

Energy or peak demand savings under a maximum technical adoption scenario for individual buildings and technologies can be assessed through many traditional activities, studies, modeling tools, and automated workflows (Brøgger & Wittchen, 2018; Cerezo et al., 2017; Cerezo Davila et al., 2016; Frayssinet et al., 2018). However, sample sizes for building-specific empirical validation field trials remain

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small due to cost or real-world limitations. Urban-scale modeling typically resorts to develop prototypical building models with multipliers (Frayssinet et al., 2018; Hong, Chen et al., 2020; Li et al., 2017). Since buildings typically are not constructed to an engineering specification, performance of a technology can vary significantly based on building characteristics and local weather. The trade-offs between sample size and aggregated accuracy of modeled vs. actual building energy use, or predicted savings of a specific technology, often remains poorly characterized. The effort to create a model of every U.S. building leveraged DOE's flagship whole-building simulation engine, EnergyPlus (Crawley et al., 2001), and its companion middleware, OpenStudio (Guglielmetti et al., 2011), which support detailed thermal, lighting, and HVAC system modeling. While these tools have enabled detailed physics-based simulations for individual buildings, applying them at national scale requires automated model generation, standardized assumptions, and computationally efficient workflows, creating significant data, computational, and methodological challenges.

To address this gap, we present *AutoBEM*, an integrated computational framework capable of generating, simulating, and evaluating energy models for nearly every building in the United States. Built on the Model America dataset and powered by high-performance computing, *AutoBEM* processes diverse geospatial inputs, including satellite, aerial, and street-level imagery, to automatically generate and simulate over one million building models per hour. This national-scale effort consumed over 45 million supercomputing core-hours and produced EnergyPlus and OpenStudio simulations for 122.9 million buildings, representing 97.8% of the U.S. building stock (J. New et al., 2021).

AutoBEM fills a critical methodological and operational gap in the energy modeling community by combining detailed physics-based simulation, large-scale automation, and actionable retrofit analytics. To support transparency and broader adoption, building-level metadata, including geometry, usage type, climate zone, and simulation results, have been made publicly accessible for 125.7 million structures in Model America v1.0 (MAv1) dataset (Bass, New, Berres, Clinton et al., 2022). Future releases under Model America v2.0 (MAv2) will incorporate additional buildings, refined input layers, and enhanced modeling workflows (Bass, New, Berres, Clinton et al., 2022).

Accordingly, *AutoBEM* is intended primarily as a scalable decision-support and research platform for stock-level analysis, retrofit screening, emissions assessment, and climate-scenario evaluation, rather than as a substitute for building-specific audits or detailed engineering design.

1.1. Empirical validation and interpretation of model reliability

AutoBEM has been applied across city, county, utility, and national scales to support building energy modeling analyses addressing electricity-saving technologies, peak-demand reduction strategies, climate change projections, and related applications (Bass & New, 2020; Bass, New, Berres, 2022; Bass, New, Copeland, 2021). Because *AutoBEM* is designed for scalable urban building energy modeling rather than manually calibrated engineering simulation of each individual building, its validation must be interpreted across multiple levels: (1) validation of input building-stock attributes, (2) individual-building energy validation, and (3) aggregate stock-level validation.

Empirical validation remains a critical challenge in the broader UBEM literature. Robust validation is needed to establish best practices, quantify uncertainty, and define appropriate use cases for model outputs. Many UBEM studies rely on fragmented or non-scalable datasets, such as local tax assessor records (New et al., 2020), and often include relatively small samples of buildings, with reported individual-building errors spanning wide ranges (Oraiopoulos & Howard, 2022). In this context, the validation evidence for *AutoBEM* and Model America is a combination of building-attribute checks, comparisons with metered energy data, and aggregate stock-level benchmarking. To make the validation evidence more transparent, Table 1 organizes prior validation studies by validation level, dataset, scale, key findings, and implications for model use, distinguishing individual-building uncertainty from aggregate stock-level reliability.

The earliest large-scale validation efforts were conducted in partnership with the Electric Power Board (EPB) of Chattanooga, Tennessee, USA. These studies compared building-level simulation results with 15-minute interval, whole-building electricity use data from large samples of buildings (New, Adams, Garrison et al., 2018). This process required detailed spatial mapping of Global Positioning System (GPS) coordinates linking advanced metering infrastructure elements to corresponding buildings, along with rigorous quality assurance and control measures (Garrison & New, 2021). Seventeen data sources were analyzed, multiple predictive techniques were developed, including inferring building type from energy signatures (Bass, New, Ezell et al., 2021), and modeling algorithms were iteratively improved to reduce discrepancies between simulated and measured data.

At the individual-building level, validation results show substantial uncertainty, as expected for uncalibrated UBEM. EPB's service territory covers approximately 1400 km^2 across eight counties centered on Chattanooga, Tennessee. Using 2019 15-minute AMI data for 50,843 buildings, uncalibrated *AutoBEM* simulations differed from measured electricity use by a median of −32% across buildings; the middle 50% of building-level errors, expressed as the interquartile range, extended from −91% to 12%, with $R^2 \approx 0.07$ and a best-fit slope of approximately 0.75 (Bass, New, Clinton et al., 2022). After simple bias adjustments approximating conditioned-area fractions and non-electric heating, the median error improved to −6%, the interquartile range shifted to −54% to 30%, and the slope increased to approximately 0.90 (Bass, New, Clinton et al., 2022). Errors were largest in winter months, consistent with uncertainty in heating fuels, partial conditioning, and other building-specific operational characteristics not directly observed in large-scale input datasets.

Validation of Model America input fields has also been conducted using independent datasets. Comparisons have included LA100 (Cochran et al., 2021), building heights in Las Vegas (New, Bass et al., 2021), and local tax assessor datasets for cities such as Flagstaff, Arizona (New et al., 2023). For example, analysis of Las Vegas building height data showed that the tallest structure in MAv1, estimated at 163 m, corresponded to the location of the Wynn Hotel, whose official height is 187 m. Of the 589,586 buildings detected in Las Vegas, MAv1 identified 324,004 buildings, or 55%, as under 5 m, primarily residential structures, while approximately 1%, or 6354 buildings, were unrealistically below 1 m. These anomalies were spatially clustered in vegetated or infrastructure-dense areas, such as Las Vegas Wash, indicating systematic challenges in height estimation due to natural obstructions and surrounding built features. These findings informed subsequent improvements in MAv2, including enhanced building height estimation and building type classification.

At aggregated scales, validation results show substantially stronger agreement. More recent comparisons of MAv2 models against annual electricity and natural gas use across five cities yielded an average total energy error of 2.3% (Bass, New, Berres, Adams et al., 2022). A comprehensive city-scale empirical validation of UBEM was also performed for Chicago, Illinois, USA (Garg et al., 2024). This study examined *AutoBEM* simulations for 596,064 buildings in the MAv2 dataset against metered electricity consumption data from 247,188 buildings averaged across 2019–2021. The analysis also incorporated comparisons with tax assessor data for 579,829 buildings, tax assessor data supplemented with MAv2, and 102 dynamic representative archetypes. Results demonstrated strong agreement between modeled and aggregated metered electricity data across multiple building types and vintages, with an initial normalized mean bias error (NMBE) of 10.9% and 1.1% after outlier removal. The study further identified building type, land use, construction age, and size as statistically significant drivers of UBEM bias, providing guidance for future calibration and standardization.

Table 1
Summary of *AutoBEM* and model America validation evidence across scales and use cases.

Validation level	Dataset/benchmark	Scale	Key findings	Interpretation for model use
Input building attributes	LA100, local tax assessor records, Las Vegas height data, Flagstaff building datasets (Cochran et al., 2021; New, Bass et al., 2021; New et al., 2023)	City- and dataset-specific comparisons	Model America input fields have been compared against independent benchmarks for geometry, height, building type, and related attributes. Las Vegas height validation identified generally plausible high-rise estimates but also systematic errors for a small subset of very low-height buildings in vegetated or infrastructure-dense areas.	Supports use of Model America for large-scale building-stock characterization, while indicating that local input errors may affect individual-building results and should be considered in detailed applications.
Individual-building energy validation	EPB 15-minute AMI electricity data (Bass, New, Clinton et al., 2022; Garrison & New, 2021; New, Adams, Garrison et al., 2018)	50,843 buildings in the EPB service territory	Uncalibrated annual electricity simulations showed a median building-level error of −32%, with an interquartile range from −91% to 12% and $R^2 \approx 0.07$. After bias adjustments for conditioned-area fractions and non-electric heating, the median error improved to −6%, the interquartile range shifted to −54% to 30%, and the best-fit slope increased to approximately 0.90.	Individual-building predictions from uncalibrated models should be interpreted as approximate engineering estimates suitable for screening and prioritization, not as calibrated utility-bill predictions without additional building-specific data.
Aggregate city/stock validation	Annual electricity and gas comparisons across five cities (Bass, New, Berres, Adams et al., 2022)	City-scale aggregate energy use	MAv2 comparisons against annual electricity and gas use across five cities yielded an average total energy error of 2.3%.	Aggregate outputs are more reliable than individual-building estimates because building-level errors partially offset across large samples. Suitable for city-scale energy and emissions planning.
Grouped UBEM validation	Chicago metered electricity data and tax assessor comparisons (Garg et al., 2024)	596,064 modeled buildings; 247,188 metered buildings	Chicago validation showed an initial NMBE of 10.9% and 1.1% after outlier removal when comparing modeled and metered electricity consumption across aggregated building groups. Building type, land use, construction age, and size were identified as significant drivers of UBEM bias.	Supports use of *AutoBEM* for building-group, neighborhood, and city-scale comparative analysis. Also identifies key bias drivers that can guide calibration and uncertainty reduction.

Taken together, these validation studies indicate that confidence in *AutoBEM* outputs depends strongly on the intended scale and application. At the individual-building level, uncalibrated outputs should be treated as approximate engineering estimates suitable for screening, prioritization, and comparative analysis rather than as calibrated predictions of actual utility bills. At aggregated spatial scales, such as neighborhoods, cities, utility territories, or building type and vintage groups, the framework provides substantially more reliable estimates because individual-building uncertainties partially offset one another. Therefore, the results in this study are interpreted primarily as stock-level and comparative planning outputs, including spatial prioritization, emissions assessment, climate-scenario comparison, and retrofit screening, rather than as investment-grade predictions for any single building without additional calibration.

1.2. Contributions and innovations

One of the key innovations of *AutoBEM* lies in the development of a scalable and modular framework for automated, physics-based urban building energy modeling at the national scale. Rather than relying solely on representative stock sampling approaches, *AutoBEM* automates the generation and simulation of building-resolved EnergyPlus/OpenStudio models using GIS-derived geometry, probabilistic building-type assignments, prototype-based operational assumptions, and location-specific climate and utility conditions. This enables scalable urban building energy modeling while balancing modeling fidelity and computational tractability.

Like many large-scale UBEM frameworks, *AutoBEM* incorporates standardized assumptions and generalized operational parameters to enable simulation across millions of buildings. The framework is therefore intended for scalable comparative analysis, retrofit screening, and stock-level planning rather than detailed engineering-grade calibration of every individual building. Nevertheless, *AutoBEM* preserves individualized building geometry, climate location, construction vintage, and building-type-specific simulation workflows, distinguishing it from purely archetype-based stock models.

Moreover, the software architecture is designed to support extensibility, with the potential to evolve into a general-purpose software development kit (SDK) for the broader UBEM community. This opens opportunities for researchers and developers to build custom workflows, plug-ins, calibration pipelines, and large-scale retrofit analysis tools using the *AutoBEM* infrastructure.

We further extend the core simulation capabilities of *AutoBEM* with several integrated analysis workflows: (1) quantification of both CO_2-equivalent and post-processed anthropogenic heat emissions (AHE) at the building level; (2) scalable evaluation of 151 energy conservation measures (ECMs) using localized utility pricing, autosized retrofit configurations, and archetype-based acceleration strategies; and (3) integration of Future Typical Meteorological Year (fTMY) weather scenarios to assess long-term climate impacts and climate-sensitive retrofit performance.

This paper outlines the structure and capabilities of the *AutoBEM* framework, describes the automated modeling workflow and data sources, compares the framework with existing UBEM platforms and stock-modeling approaches, and presents a Phoenix, Arizona case study demonstrating scalable, data-driven retrofit and decarbonization analysis.

The primary intended users of *AutoBEM* are researchers, public-sector planners, policymakers, utilities, and energy-program administrators who require scalable, building-resolved information for stock-level analysis and decision support. Core use cases include baseline energy and emissions assessment, spatial prioritization of high-impact building segments, screening of retrofit measures, evaluation of climate-sensitive energy demand, and comparison of policy or program scenarios. Practitioners such as engineers, auditors, and retrofit providers may use *AutoBEM* outputs for early-stage portfolio screening, but project-level design, financing, and implementation decisions require additional building-specific data, calibration, engineering review, and local cost estimates.

2. Research gaps and innovations

Despite significant progress, current UBEM tools face three limitations when used for screening- and planning-level retrofitting at scale: (1) *Coverage:* most platforms operate at city or regional scope or rely on sampling-based archetypes (e.g., ComStock/ResStock), limiting building-level fidelity across the national stock; (2) *Integrated ECM evaluation:* ECM analyses are typically decoupled from autosized equipment costs, localized utility prices, and CO_2 emission assessment accounting within one automated workflow; (3) *Climate-forward assessment:* few pipelines quantify ECM performance under future weather (fTMY) in a way that is reproducible and scalable to tens of millions of buildings.

To address these gaps, *AutoBEM* contributes a unified and scalable pipeline that:

(a) generates *individualized*, physics-based models for 122.9 M U.S. buildings (97.8% coverage) using GIS-derived geometry and DOE-consistent prototypes;
(b) couples building-level *ECM evaluation* (151 measures) with *autosized HVAC capacities* and a *cost engine* (component-level $/kW or $/unit) plus *location-specific* utility pricing;
(c) evaluates *CO_2-equivalent* emissions using subregion eGRID factors, combining CO_2, CH_4, and N_2O with stated GWPs;
(d) integrates *fTMY* weather to assess climate-robust retrofit value across multiple SSP–RCP futures;
(e) releases public, building-level inputs and outputs (MAv1) to support reproducibility, benchmarking, and downstream research.

This framework yields (i) policy-relevant scale through national coverage at building resolution; (ii) traceability through open inputs, outputs, and reproducible workflows; and (iii) actionable screening metrics, including costed payback and emissions impacts under current and future climate conditions.

3. Related work

UBEM tools and building-stock modeling approaches have expanded substantially in recent years. Existing methods span a wide spectrum, including physics-based urban simulation platforms, district and campus energy-system tools, reduced-order stock models, GIS-to-BEM translation workflows, and national-scale representative-building datasets (Ferrando et al., 2020). Recent reviews emphasize that UBEM tool selection depends strongly on the intended use case, required input data, spatial scale, computational burden, validation needs, and target users (Ferrando et al., 2020; Kong et al., 2023). The comparison in this section is not intended to imply that all tools are identical in function or scope, but rather to position *AutoBEM* within the broader landscape of scalable building energy modeling approaches.

A substantial body of prior work has used archetype-based or representative-building approaches for large-scale stock assessment. These studies typically aggregate the building stock by building type, age, climate, and system characteristics, enabling evaluation of energy conservation measures and policy impacts without generating a unique simulation model for every individual structure. For example, Mata et al. used a harmonized bottom-up building stock modeling methodology to compare retrofit energy savings, CO_2 mitigation, and cost-effectiveness across five EU member states (Mata et al., 2018). Similarly, Luddeni et al. (2018) developed a bottom-up methodology for large-scale deep retrofit analysis of the Italian office building stock, using reference buildings, EnergyPlus simulation, cost-optimal optimization, and representation indices to scale retrofit results from individual prototypes to the national stock. Such approaches have been highly influential for policy analysis and retrofit planning, particularly where individualized building-level inputs are unavailable.

National-scale U.S. stock models such as ComStock and ResStock extend this representative-building approach to the commercial and residential sectors, respectively (Parker et al., 2023; Reyna et al., 2022). These frameworks leverage national building-stock datasets, prototype assumptions, and statistical sampling to evaluate energy performance, energy conservation measures, and policy scenarios. Their strength lies in national representativeness and scenario analysis, but they generally do not generate individualized geometry-based simulation models for every physical building. Recent work has also explored machine-learning-assisted approaches that combine national stock datasets with GIS-derived geometry to reduce dependence on detailed building metadata. Wan and Meyer (2025), for example, matched real buildings to ComStock and ResStock records using location and geometric similarity, then trained Random Forest models to predict EUI for a Chicago case study.

In parallel, several UBEM platforms have been developed to support urban, district, or regional simulation using GIS, CityGML, OpenStreetMap, or other spatial datasets. For example, CityBES uses urban geometry and prototype building templates to support EnergyPlus-based city-scale simulation and retrofit analysis (Chen et al., 2017). SimStock translates GIS-derived building data into stock-scale energy models, with particular emphasis on UK and European applications (University College London (UCL), 2025). More recently, Zhang et al. (2025) proposed GeoBEM, a geospatial-computing-enabled UBEM framework that integrates multi-source GIS data fusion, GIS-based thermal zoning, terrain-aware shading modeling, and prototype building models for city-scale energy simulation. Other tools, such as SimStadt, TEASER, the City Energy Analyst (CEA), and URBANopt, provide important capabilities for urban energy-system analysis, reduced-order stock modeling, district-scale scenario evaluation, or community energy planning (El Kontar et al., 2020; Fonseca et al., 2016; Remmen et al., 2018; Weiler et al., 2021). These tools demonstrate the value of geometry-informed UBEM, but many are primarily designed for city, district, campus, or regional applications and may rely on detailed local datasets that are not consistently available at national scale.

Table 2 summarizes representative UBEM platforms and large-scale building-stock modeling approaches relevant to this study (Chen et al., 2017; Mata et al., 2018; New, Adams et al., 2021; New, Adams, Im et al., 2018; Parker et al., 2023; Reyna et al., 2022; University College London (UCL), 2025). The table distinguishes among urban-scale simulation platforms, district or community energy tools, reduced-order modeling frameworks, and national stock models. These approaches are not directly equivalent, but they provide useful reference points for understanding the methodological position of *AutoBEM*.

AutoBEM, developed by Oak Ridge National Laboratory (ORNL), is positioned at the intersection of these traditions. Similar to geometry-based UBEM platforms, it supports automated generation and simulation of building-resolved EnergyPlus/OpenStudio models using GIS-derived building information. Similar to national stock-modeling frameworks, it also supports scalable analysis across very large building populations, using archetype aggregation and scaling methods for computationally efficient retrofit evaluation. Its primary contribution is the integration of national-scale building-resolved geometry, automated physics-based simulation workflows, building-level baseline outputs, scalable ECM screening, CO_2-equivalent emissions, post-processed anthropogenic heat emissions, and fTMY weather integration within one framework.

This positioning addresses a gap between city-scale UBEM platforms and national representative-building stock models. City- and district-scale tools often provide detailed geometry-informed workflows but are limited in geographic scope or data availability, while national stock models provide broad coverage but typically rely on representative sampling rather than individualized geometry for each building. *AutoBEM* links the Model America building-stock dataset with automated EnergyPlus/OpenStudio simulation and scalable decision-support workflows, enabling building-resolved national-scale analysis while maintaining computational tractability through standardized assumptions and archetype-based acceleration where needed.

Table 2
Representative UBEM platforms and building-stock modeling approaches relevant to scalable building energy analysis.

Tool/framework	Developer	Primary category	Typical scale	Modeling approach	Geometry/input basis	Key capabilities	Limitations relative to this study
AutoBEM	ORNL	Automated UBEM/national stock modeling	City to national	Building-resolved EnergyPlus/OpenStudio simulations with scalable assumptions	Model America, GIS-derived geometry, inferred type, height, and vintage	National-scale automated model generation; baseline energy; CO_2; post-processed AHE; ECM screening; fTMY integration	Single-zone national implementation; uncertainty from inferred inputs; ECM results use archetype-based acceleration in this study
CityBES	LBNL	Urban-scale UBEM platform	City/district	EnergyPlus-based UBEM	CityGML/urban geometry and building-stock data	City-scale energy modeling, retrofit analysis, and visualization	Primarily city-scale; dependent on local data availability; not designed as a full national building-resolved stock dataset
URBANopt	NRL	District/campus simulation SDK	District/campus/ community	OpenStudio/EnergyPlus-based scenario modeling with district energy and grid integration	Building footprints, heights, types, district energy-system inputs	District-scale scenario analysis, DER analysis, district thermal systems, grid-interactive buildings	Focused on district/community energy systems rather than nationwide building-stock simulation
TEASER	RWTH Aachen	Reduced-order UBEM/stock modeling tool	District/city/regional	Reduced-order thermal models for building stocks	GIS, CityGML, archetype and building-stock data	Automated model generation and reduced-order simulation for urban energy studies	Primarily reduced-order modeling; not focused on U.S. national building-resolved EnergyPlus simulations
SimStadt	HFT Stuttgart	GIS-based urban energy platform	District/city/regional	CityGML-based urban energy analysis	CityGML and 3D city models	Heating demand, PV potential, refurbishment, and renewable energy scenarios	Requires structured 3D city data; primarily regional/city applications, especially European contexts
City Energy Analyst (CEA)	ETH Zurich/ collaborators	Urban energy systems analysis platform	District/city	Urban energy demand and energy-system optimization	GIS inputs, building properties, schedules, archetypes	Urban energy demand, supply-system optimization, district energy planning	More focused on district/city energy-system planning than national-scale building-resolved stock simulation
SimStock	UCL	GIS-to-BEM/stock modeling	City/regional	GIS-derived stock modeling and simulation	Building footprints and UK/EU building-stock data	Large-scale stock modeling, especially for UK/EU contexts	Regional focus; not designed for full U.S. building-stock simulation
ComStock	NRL	National commercial stock model	National	Representative-building sampling/stock modeling	Commercial building stock datasets and prototype assumptions	National commercial building stock analysis, ECM and policy scenarios	Commercial only; sampling-based; no individualized geometry for every building
ResStock	NRL	National residential stock model	National	Representative-building sampling/stock modeling	Residential housing stock data and prototype assumptions	National residential stock analysis, ECM and policy scenarios	Residential only; sampling-based; no individualized geometry for every building

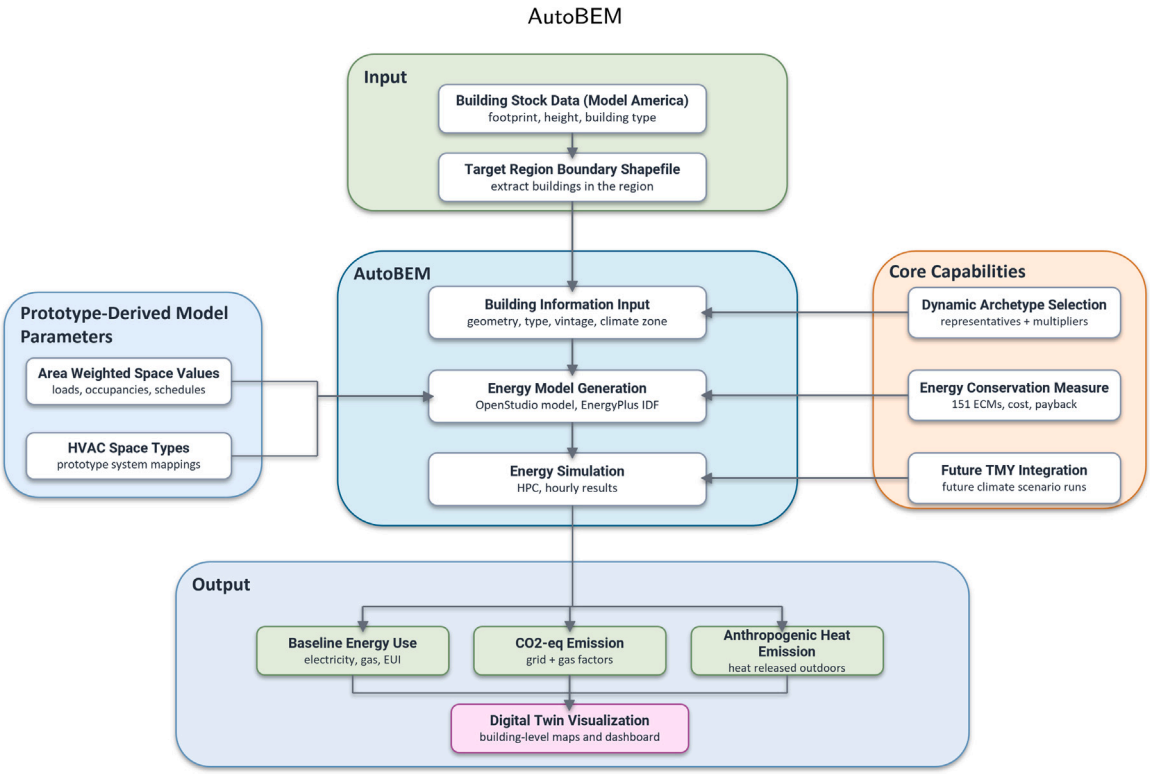


Fig. 1. Overall framework of *AutoBEM* illustrating the integrated workflow from Model America data inputs to digital twin visualization, with core and additional capabilities for urban energy simulation and retrofit evaluation.

Taken together, the literature reflects a spectrum of UBEM approaches. At one end are archetype-based stock models, which are well suited to national or cross-country policy analysis but generally lack individualized building geometry. At the other are city-scale or district-scale UBEM platforms that incorporate GIS-derived geometry and detailed simulation workflows but are often limited in geographic scope. National U.S. tools such as ComStock and ResStock provide broad stock representation through sampling, while tools such as CityBES and SimStock emphasize urban-scale geometric modeling. *AutoBEM* is positioned at the intersection of these traditions by combining national-scale coverage with individualized, physics-based building models, building-level ECM evaluation, and climate-forward analysis.

4. Methodology

The *AutoBEM* platform provides a comprehensive and scalable approach for building energy modeling across urban areas. Fig. 1 illustrates the framework of *AutoBEM*. It begins with data acquisition, utilizing the Model America building stock dataset and GIS shapefiles to extract buildings in the target regions. These data serve as input for generating detailed energy models. The platform integrates core capabilities such as dynamic archetype selection for representative building modeling, targeted retrofit measure selection, and implementation strategies, as well as predictive simulations using fTMY weather data. Simulation outputs provide essential insights based on baseline energy consumption (i.e., energy use of the existing building configuration prior to any retrofit intervention), CO_2-equivalent emissions, and post-processed anthropogenic heat emission assessments, ultimately enabling comprehensive urban digital twin visualizations for informed decision-making and effective urban energy planning.

4.1. Input data

The *AutoBEM* framework relies on the *Model America* dataset, a high-resolution, nationwide compilation of building-level geospatial and physical attributes covering over 125 million U.S. buildings. This dataset was developed using open-access geospatial data and advanced remote sensing methods to generate consistent input parameters for energy modeling workflows. It integrates diverse data sources, including satellite, aerial, and street-view imagery; LiDAR-based elevation maps; building codes; engineering heuristics; artificial intelligence; and other big data analytics to capture detailed building characteristics. Each building model incorporates between 281 and 4617 attributes describing its geometry and contextual environment. While internal characteristics — such as envelope thermal properties, HVAC configuration, and occupancy schedules — are based on prototypical building templates aligned with prevailing building codes and construction periods, the foundational external inputs include building footprints, heights, building type, and building vintage. These comprehensive inputs enable automated generation of physics-based energy models using OpenStudio and EnergyPlus.

Table 3 summarizes the data fields developed through this process, which collectively support automated model generation for over 125 million U.S. buildings. These fields include geospatial metadata, physical geometry, typology, energy performance characteristics, and simulation results.

Building footprints. Building geometries were sourced from Microsoft's national building footprint dataset, which includes over 125 million buildings across all 50 states. The dataset defines each building as a 2D polygon in geographic coordinates (latitude and longitude), generated using semantic segmentation and polygonization from high-resolution imagery (Microsoft, 2020). To improve simulation reliability and performance, complex geometries were simplified into regularized polygons. A centroid was computed by averaging the footprint's latitude and longitude points. This centroid serves as the basis for generating a *Unique Building Identifier (UBID)*, a reproducible geospatial encoding scheme that facilitates spatial indexing and integration with other building datasets.

Table 3
Summary of input fields in the Model America dataset.

Field	Description
UBID	Unique geospatial identifier for spatial indexing.
Identifier	Filename reference for individual energy models.
County/State	Geographic location of the building.
CZ /Clim_Zone	ASHRAE climate zone designation.
est_year	Estimated year of construction.
est_commercial	Binary indicator: 0 = residential, 1 = commercial.
Footprint2D	2D polygon representing the building footprint.
Centroid	Latitude and longitude of the footprint centroid.
Height (m)	Estimated vertical height of the structure (in meters).
Area2D (ft^2)	Footprint area (in square feet).
BuildingType	DOE prototype building type assignment.
WWR_surfaces	Window-to-wall ratio for each facade.
NumFloors	Estimated number of above-grade floors.
Area (ft^2)	Total conditioned floor area (in square feet).
Standard	Building code vintage or energy standard reference.

Building heights. Building height was estimated using convolutional elevation profiling applied to the *Advanced Land Observing Satellite (ALOS) World 3D 30 m (AW3D30)* digital surface model, provided by JAXA and accessed via Google Earth Engine (Tadono et al., 2016). For each structure, the local minimum elevation of surrounding terrain was subtracted from the maximum elevation to estimate vertical height, with additional slope correction following the method described in Marconcini et al. (2014). In cases where height data was unavailable, missing values were imputed using the median height of the 20 nearest neighboring buildings.

Building types. Building type assignment was determined probabilistically based on each building's estimated height and two-dimensional footprint area. First, buildings were grouped into physical bins using simple height and footprint-area thresholds. These thresholds were heuristically derived from typical physical characteristics of the DOE prototype building types, including low-rise small-footprint buildings, low-rise large-footprint buildings, mid-rise buildings, and high-rise buildings (Deru et al., 2011). Each bin was used to restrict the set of plausible DOE prototype building types that could be assigned to a building, rather than assigning a building type deterministically, as shown in Table 4.

Within each bin, the final building type was selected probabilistically. For commercial building candidates, the assignment probabilities were based on construction weight distributions from the 2003–2007 Commercial Buildings Energy Consumption Survey (CBECS) (Jarnagin & Bandyopadhyay, 2010). Because these construction weights only represent commercial buildings, residential building types were incorporated using a national residential-to-commercial ratio of 95.88% residential to 4.12% commercial, derived from the 2012 CBECS and 2015 Residential Energy Consumption Survey (RECS) (U.S. Energy Information Administration, 2015a, 2015b). The residential portion of the probability was assigned to the appropriate residential prototype candidate within each bin, including IECC low-rise residential, MidriseApartment, or HighriseApartment, while the commercial portion was distributed among the commercial prototype candidates according to the CBECS construction weights. Because the assignment is based on inferred geometry and stock-level probability weights, the resulting building type represents a scalable modeling input rather than a verified parcel-level land-use classification.

The following building types are currently supported within the *AutoBEM* framework, aligned with DOE prototype classifications: *SmallOffice*, *MediumOffice*, *LargeOffice*, *Warehouse*, *RetailStandalone*, *RetailStripmall*, *QuickServiceRestaurant*, *FullServiceRestaurant*, *Supermarket*, *PrimarySchool*, *SecondarySchool*, *OutpatientHealthcare*, *Hospital*, *SmallHotel*, *LargeHotel*, *MidriseApartment*, *HighriseApartment*, and *IECC*. The *IECC* building type refers to detached low-rise residential buildings (1–2 stories) that conform to the International Energy Conservation Code (IECC), a model code widely adopted across U.S. states to regulate residential building energy efficiency. The resulting building type classification was then used to assign prototype-based modeling inputs, including floor-to-floor height, window-to-wall ratio, schedules, internal loads, ventilation assumptions, and HVAC system templates. Floor counts were computed by dividing the estimated building height by the floor-to-floor height associated with the assigned prototype.

Building vintages. Construction year was estimated using temporal classification of urbanization patterns based on the *Global Artificial Impervious Area (GAIA)* dataset at 30-meter spatial resolution, which identifies the first year a surface was urbanized (post-1985). This year was used as a proxy for construction vintage. Gaps in the dataset were filled using the most frequent vintage among neighboring structures. For buildings where vintage information was still unavailable, fallback assignments were made using national-level construction year distributions from RECS and CBECS surveys (U.S. Energy Information Administration, 2015a, 2015b). Buildings were then categorized into standardized code vintages, aligning with building code eras (e.g., Pre-1980,1980–2004, 2004–2012, Post-2012).

4.1.1. Prototype-derived model parameters

To account for variations in internal building characteristics while maintaining national-scale computational feasibility, *AutoBEM* uses prototype-derived model parameters that are not directly obtained from national geospatial datasets or Model America. These parameters are derived from prototype libraries for each DOE building type, building standard (or construction vintage), and ASHRAE climate zone combination using the area-weighting approach (National Laboratory Rockies, 2026). The retrieved parameter set includes prototype-derived assumptions for occupancy density, electric and gas equipment loads, interior lighting, exterior lighting and equipment loads, service water heating, ventilation, parasitic loads, and internal heat-gain fractions. These quantities are derived from DOE prototype building definitions and corresponding OpenStudio/EnergyPlus model assumptions (Deru et al., 2011; National Laboratory Rockies, 2026). Table 5 summarizes the major parameter categories used in the *AutoBEM* model-generation workflow. For each building, the assigned building type, standard, and climate zone are used to retrieve the corresponding parameter set. For example, the model parameters for a building classified as LargeOffice, assigned to ASHRAE 90.1–2013, and located in climate zone 1A, are shown in Table 5.

Because the national-scale implementation of *AutoBEM* represents each building as a single thermal zone, these parameters from multi-space DOE prototype buildings are converted into equivalent single-zone values using area-weighting approach. This conversion is applied to scalar load, density, and flow quantities for which an equivalent whole-building value is physically meaningful, as listed in Table 5. These area-weighted values are intended to preserve the aggregate internal-load, ventilation, and service-load characteristics of the corresponding DOE prototype building when represented as a single-zone energy model.

Because the current national-scale implementation of *AutoBEM* treats each building as a single thermal zone, these calibrated prototype-derived model parameters serve as approximations of space-type characteristics within each building. This approach enables consistent and computationally tractable simulation across millions of buildings, while preserving alignment with DOE prototype-based assumptions. However, it also introduces uncertainty by aggregating spatially heterogeneous conditions into a simplified representation.

In particular, the use of prototype-derived parameters in single-zone models can smooth differences among functional areas within a building, including variations in internal loads, occupancy, solar exposure, ventilation requirements, and HVAC operation. Recent studies have shown that simplified zoning or parameter aggregation may lead to cancellation of opposing thermal loads, such as simultaneous

Table 4
Height- and footprint-area bins used to restrict candidate DOE prototype building types before probabilistic assignment.

Physical bin	Candidate prototype building types
Height < 40 ft and footprint area < 6000 ft^2	FullServiceRestaurant; QuickServiceRestaurant; SmallOffice; IECC low-rise residential
Height < 40 ft and footprint area 6000–15,000 ft^2	RetailStandalone; RetailStripmall; OutpatientHealthcare
Height < 40 ft and footprint area > 15,000 ft^2	PrimarySchool; SecondarySchool; Warehouse
Height 40–80 ft and footprint area < 18,000 ft^2	MediumOffice; SmallHotel; MidriseApartment
Height 40–80 ft and footprint area > 18,000 ft^2	Hospital; LargeHotel
Height > 80 ft	HighriseApartment; LargeOffice

Table 5
Summary of prototype-derived model parameters used in AutoBEM, with example values shown for a LargeOffice building assigned to ASHRAE 90.1–2013 in climate zone 1A.

Parameter category	Parameters	Description	Example: LargeOffice/90.1–2013/1A
Occupancy	People density	Occupant density used to represent people-related internal gains and occupancy-driven loads in the single-zone model.	0.079022 people/m^2
Interior electric loads	Electric equipment load; interior lighting load	Prototype-derived electric equipment and lighting loads represented as equivalent single-zone scalar values.	Electric equipment: 25.5089 W/m^2; interior lighting: 8.8857 W/m^2
Interior gas equipment	Gas equipment load; gas equipment availability	Prototype-derived gas equipment load where applicable. Gas availability is a yes/no assumption indicating whether gas equipment is present.	Gas equipment: not applicable; gas availability: No, unitless
Exterior and parasitic loads	Exterior lighting load; exterior equipment load; parasitic load	Prototype-derived loads associated with exterior lighting, exterior equipment, and auxiliary or parasitic energy uses.	Exterior lighting: 32,054.4 W; exterior equipment: 127,828.5 W; parasitic load: 19.9992 W
Service water heating	Service water heating load; water heater capacity; water flow rate; water-use cycle coefficient	Prototype-derived parameters used to represent service hot-water demand, water heater sizing, water use rate, and temporal water-use behavior.	Service water heating load: 1.1356 W/m^2; water heater capacity: 87,921.3 W; water flow rate: 3.66×10^{-5} m^3/s; water-use cycle coefficient: 5.9980
Ventilation	Outdoor air requirement	Prototype-derived outdoor air requirement converted to an equivalent single-zone ventilation value.	0.00042846 $m^3/(s\ m^2)$
Electric equipment heat-gain fractions	Radiant fraction; latent fraction; lost fraction for electric equipment	Fractions describing how electric equipment heat gains are distributed to the thermal zone, latent load, or losses outside the zone.	Radiant: 0.15; latent: 0.00; lost: 0.00
Gas equipment heat-gain fractions	Radiant fraction; latent fraction; lost fraction for gas equipment	Fractions describing how gas equipment heat gains are distributed when gas equipment is present.	Radiant: 0.00; latent: 0.00; lost: 0.00

heating and cooling demands in different functional zones, potentially underestimating energy demand and peak loads (Zhou et al., 2025). These effects are expected to be more important for large, mixed-use, or geometrically complex buildings than for smaller or more homogeneous buildings. The limitation is discussed further in Section 5.6. Despite this limitation, such aggregation is adopted in *AutoBEM* to ensure computational tractability when simulating millions of buildings. The area-weighting single-zone approach represents a trade-off between model fidelity and scalability, enabling nationwide simulation while maintaining consistency with DOE prototype-based assumptions. It is most appropriate for annual energy estimation, stock-level comparison, emissions assessment, and large-scale retrofit screening. Detailed zone-level comfort analysis, equipment sizing, and peak-load evaluation would require higher-fidelity multi-zone models and additional building-specific input data.

4.1.2. HVAC systems

In addition to the prototype-derived values, *AutoBEM* integrates predefined Heating, Ventilation, and Air Conditioning (HVAC) configurations based on building types and vintages. These HVAC configurations encompass critical system details including HVAC system type (e.g., Packaged Single Zone Air Conditioner, PSZ-AC), operational schedules, outdoor air damper controls, fan operation characteristics (constant or variable volume), and detailed cooling and heating system specifications, as shown in Table 6. The standardized HVAC parameters are derived from DOE prototype models and ASHRAE standards (National Laboratory Rockies, 2026). Therefore, *AutoBEM* reflects prototype-consistent HVAC configurations, operational conditions and energy demands of buildings at the urban scale. By integrating these HVAC system types, *AutoBEM* captures the nuanced differences in HVAC performance across diverse building typologies and vintages, significantly improving the realism and applicability of simulation results for urban energy analyses.

For example, the LargeOffice template for ASHRAE 90.1–2013 assigns a variable-air-volume system with reheat, a ground-floor return plenum, a large-office HVAC operation schedule, a minimum outdoor-air motorized damper schedule, a constant-primary variable-secondary chilled-water pumping configuration, and a water-cooled centrifugal chiller, as demonstrated in Table 6. Heating and cooling equipment capacities are subsequently determined through EnergyPlus autosizing using the generated building geometry, local weather file, internal loads, ventilation assumptions, and selected HVAC configuration.

4.2. Core capabilities

4.2.1. Dynamic archetypes selection

Building archetypes are simplified, prototypical representations of real buildings that are fundamental to urban-scale energy modeling, as they balance the need for detailed physical characterization with computational tractability (Zhuang et al., 2023). According to Shen and Wang, archetypes can be developed through three main approaches (Shen & Wang, 2024). The first is the code-based approach, which

Table 6
Summary of prototype-derived HVAC parameters used in AutoBEM, with example parameters shown for the LargeOffice template under ASHRAE 90.1–2013.

HVAC template parameter	Description	Example: LargeOffice/90.1–2013
HVAC system type	General category of HVAC system assigned to the building, such as variable-air-volume, packaged single-zone, or other prototype-consistent system types.	Variable air volume (VAV)
HVAC template name	Specific prototype HVAC configuration used to generate the system in the OpenStudio/EnergyPlus model.	VAV with reheat
Return plenum	Return-air plenum assignment used in the HVAC system configuration, where applicable.	Ground-floor return plenum
HVAC operation schedule	Schedule defining when the HVAC system is available or operating.	Large-office HVAC operation schedule
Outdoor-air damper schedule	Schedule controlling minimum outdoor-air damper operation.	Large-office minimum outdoor-air motorized damper schedule
Fan operation assumption	Template-based fan operation logic, such as cycling or continuous operation, where applicable.	Not specified in this template field
Chilled-water pumping configuration	Prototype-based chilled-water pumping arrangement, such as constant-primary variable-secondary pumping.	Constant-primary variable-secondary pumping
Chiller cooling type	Cooling or heat-rejection type associated with the chiller system, such as water-cooled operation.	Water-cooled chiller
Chiller condenser type	Condenser configuration assigned to the chiller system, where applicable.	Not applicable/not specified
Chiller compressor type	Compressor technology assigned to the chiller system, such as centrifugal compression.	Centrifugal compressor

relies on standard prototype definitions drawn from regulations and design guidelines. This method offers clear, replicable baselines but may overlook local construction idiosyncrasies. The second is the data-driven approach, which uses measured consumption data and clustering techniques to derive archetypes that closely reflect observed usage patterns, but requires extensive metered datasets. The third is the hybrid approach, which couples physics-based prototypes with statistical calibration, seeking to balance physical realism with empirical accuracy. By grouping thousands of heterogeneous structures into a limited set of archetypes defined by key attributes (e.g., construction vintage, size, envelope properties, HVAC systems), modelers can estimate citywide energy use without simulating every individual building (Ang et al., 2023). Shen and Wang (2024) also note that the choice among these approaches depends on research objectives, data availability, computational resources, and the desired trade-off between accuracy and scalability. Such aggregation reduces data requirements and simulation times while preserving enough diversity to capture broad patterns of energy demand (Giuzio et al., 2025).

Archetypes ensure that the full spectrum of building stock variability, such as differences in floor area, insulation levels, glazing ratios, and occupancy schedules, is reflected in aggregated energy forecasts (Wahi et al., 2024). For example, a model may include separate archetypes for pre-1970 low-rise apartments, post-2000 high-rise offices, and mid-century detached homes, each calibrated to measured or survey data (Ma et al., 2025). This stratification improves the accuracy of baseline energy assessments and sensitivity analyses by aligning simulated energy use with observed consumption trends across building categories (Ali et al., 2024).

AutoBEM generates and simulates individual building-level models for baseline energy, emissions, and anthropogenic heat analysis. However, applying all retrofit measures to every building would require a much larger number of simulations. Therefore, dynamic archetype selection is used as a computational acceleration step for large-scale retrofit evaluation.

The dynamic archetype workflow consists of four steps, as illustrated in Fig. 2. First, buildings are grouped using attributes that strongly influence energy performance and retrofit response, including building type, construction standard or vintage, climate zone, and study region. Second, one representative building is selected from each group. In this study, the representative building is the building whose floor area is closest to the median floor area of that group. This approach preserves an actual building geometry from the dataset, rather than creating an abstract average building. Third, the selected representative building is simulated under the baseline condition and under each applicable retrofit measure. Fourth, the energy savings, cost savings, CO_2-equivalent emissions reductions, and investment costs from the representative building are scaled to the full group using an area multiplier.

4.2.2. Energy conservation measure evaluation

Energy Conservation Measure (ECM) integrations are critical to *AutoBEM* for systematic evaluation of retrofit options at scale. Integrating ECMs within *AutoBEM* allows stakeholders to quantitatively compare various retrofit strategies based on their energy-saving potential, emissions reductions, cost-effectiveness, and payback periods. By integrating comprehensive ECM libraries directly into modeling workflows, *AutoBEM* can facilitate informed decision-making by urban planners, policymakers, utilities, and building owners. This integration supports the prioritization of targeted interventions across large portfolios of buildings, ultimately accelerating progress toward urban energy efficiency, decarbonization, and climate resilience goals.

AutoBEM currently supports 151 retrofitting measures, including 129 HVAC-related measures and 22 non-HVAC measures (e.g., wall insulation, roof insulation, window upgrades, and lighting retrofits). Table 7 lists representative examples of ECMs and key parameters. HVAC-related measures encompass a wide range of system configurations, including variable air volume (VAV) systems, packaged units (PSZ/PSZ-AC), dedicated outdoor air systems (DOAS), fan coil units (FCU), water-source heat pumps (WSHP), and residential system variants. These systems are paired with different heating technologies, such as gas boilers, electric resistance heating, and heat pump systems. Non-HVAC measures focus on improvements to building envelope performance and lighting efficiency. Envelope upgrades include wall and roof insulation options with varying thermal resistance levels (R-values) reflecting different materials and construction practices, while window retrofits consider a range of U-values and solar heat gain coefficients. Lighting measures include high-efficiency LED upgrades that reduce lighting power density (LPD).

Fig. 3 illustrates the workflow for evaluating retrofit measures within the *AutoBEM* framework. The process begins with the collection of measure-specific information, such as equipment costs and performance characteristics, derived from sources including National Residential Efficiency Measures Database, Facility Energy Decision System (FEDS), Google Search (National Laboratory of the Rockies (NRL), 2025; Pacific Northwest National Laboratory (PNNL), 2025). For non-HVAC measures, key geometric and material attributes (e.g., wall and

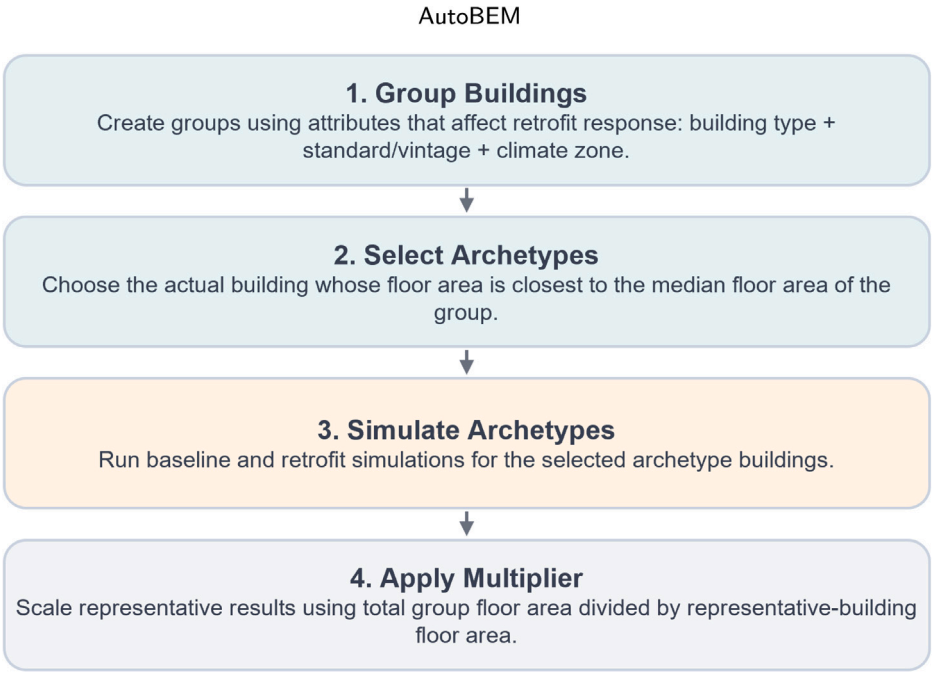


Fig. 2. Dynamic archetype workflow.

Table 7
Representative examples of ECMs and key parameters.

Type	Measure name	Description	Performance	R/U Value
Wall	R-13 Insulation	Cellulose, 2 × 4 wall	–	R-13
Wall	R-36 Insulation	Closed-cell spray foam	–	R-36
Roof	R-47 Insulation	High-performance roof	–	R-47
Window	U = 0.48	Single-pane + storm	–	U = 0.482
Window	U = 0.17	Triple-pane low-E	–	U = 0.17
Lighting	LED Retrofit	Reduced lighting power density	0.60 W/ft^2	–
HVAC	VAV + Boiler	Central VAV system	COP = 2.96	–
HVAC	Air Source Heat Pump	Electrified system	COP = 3.9	–
HVAC	VRF System	Variable refrigerant flow	COP = 4.2	–

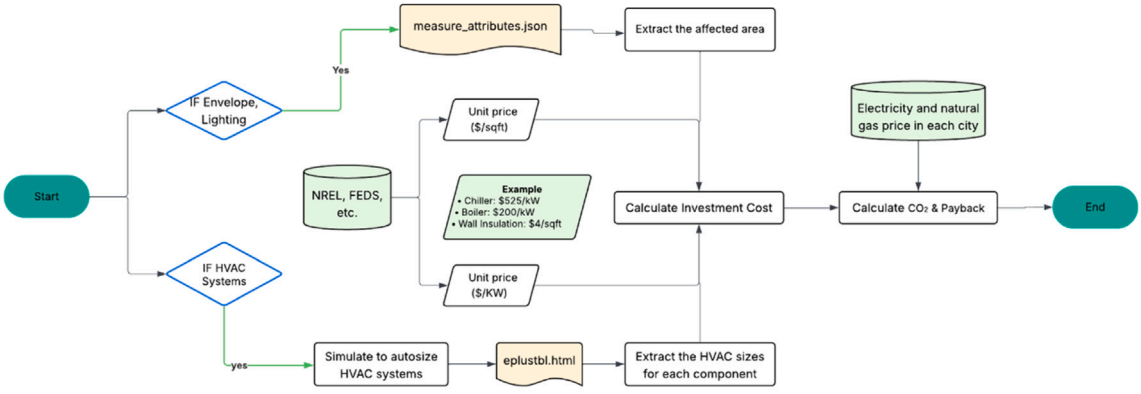


Fig. 3. Workflow for retrofitting measure evaluation.

window areas) are obtained from the `measure_attributes.json` file generated during the EnergyPlus simulation process. These attributes are subsequently used to estimate total implementation costs using standardized unit cost metrics (e.g., $/ft^2).

For HVAC retrofit measures, simulations are first conducted to automatically size system components based on building-specific heating or cooling demands. Equipment capacities such as those for cooling coils, chillers, boilers, and fans are then extracted from the EnergyPlus output report (.html). A master script was developed to store and manage unit cost information for a variety of HVAC components, including cooling coils (single-speed, variable-speed), heating coils (electric, gas, hot water), boilers (hot water and steam), chillers (air- and water-cooled), cooling towers, pumps, fans, heat exchangers, etc. Each component is assigned a standardized cost metric (e.g., $/kW or $/unit). Therefore, the total investment costs can be calculated based on the autosized

Table 8
Definitions of energy modeling output parameters used in this study.

Parameter	Definition
Electricity facility (J)	Total electrical energy consumption at the facility level.
Natural gas facility (J)	Total natural gas energy consumption at the facility level.
Heating electricity (J)	Electrical energy specifically used for heating purposes.
Cooling electricity (J)	Electrical energy specifically used for cooling purposes.
Heating natural gas (J)	Natural gas energy specifically used for heating purposes.
Heating total (J)	Aggregate heating energy consumption from all sources.
Water systems electricity (J)	Electrical energy used for water heating and pumping systems.
Lighting electricity (J)	Electrical energy used by lighting systems.
Equipment electricity (J)	Electrical energy used by equipment and appliances.
Fans electricity (J)	Electrical energy used by HVAC fans.
Pumps electricity (J)	Electrical energy used by HVAC pumps.
Heat rejection electricity (J)	Electrical energy used by heat rejection systems (e.g., cooling towers).
Heat recovery electricity (J)	Electrical energy used by heat recovery systems.

capacities. This integrated approach supports consistent, scalable evaluation of HVAC retrofit strategies across various building types and system configurations. Electricity and natural gas prices specific to each city are then incorporated to compute the economic metrics (e.g., payback) compared to the baseline model. This integrated process enables automated, location-sensitive evaluation of energy savings and cost-effectiveness for a variety of retrofit strategies.

4.2.3. Future Typical Meteorological Year (fTMY) integration

To effectively combat the impacts of climate change, global investments of approximately $1.5 trillion in building energy efficiency are projected to be necessary by the year 2030 (Kammen & Sunter, 2016). Recent studies have demonstrated that integrating future climate scenarios can reveal substantial changes in building energy demand and identify climate-specific adaptation strategies (Wang et al., 2026). Climate-responsive simulations using fTMY weather datasets were derived from Intergovernmental Panel on Climate Change (IPCC) scenario projections. It supports robust analyses at various spatial scales, from neighborhoods to the entire nation. These simulations support data-driven strategies to optimize energy use, reduce peak demand, minimize greenhouse gas emissions, and manage associated costs within the U.S. building sector.

AutoBEM can incorporate fTMY to quantify future shifts in energy consumption, emissions, and retrofit effectiveness. This capability makes *AutoBEM* suitable for evaluating long-term building performance and informing policies under climate change scenarios, complementing conventional TMY-based simulations

The fTMY files used in this study were generated by Chowdhury et al. for all 3281 U.S. counties using six IPCC climate models and the canonical TMY generation method (Chowdhury, Li, Stubbings, New, 2023). These datasets are available for four distinct Shared Socioeconomic Pathway–Representative Concentration Pathway (SSP–RCP) scenarios: SSP1–RCP2.6, SSP2–RCP4.5, SSP3–RCP7.0, and SSP5–RCP8.5. Each combination represents a different trajectory of greenhouse gas emissions and socioeconomic development, offering a comprehensive range of plausible climate futures.

The fTMY integration workflow is as follows. First, an *AutoBEM* building model is generated using the same geometry, building type, construction standard, internal loads, schedules, and HVAC assumptions used for the baseline simulation. Second, the building's location is used to select the corresponding fTMY weather file for a chosen SSP–RCP scenario and future time period. Third, the model is simulated in *AutoBEM* using the selected fTMY weather file. Finally, the future-weather simulation outputs are compared with the baseline weather simulation to quantify changes in electricity use, natural gas use, CO_2-equivalent emissions, utility cost, and retrofit performance. A case study by Li et al. found that fTMY conditions could increase building energy consumption and CO_2 emissions in the Phoenix metropolitan area by approximately 9.3% and 10.3%, respectively, using AutoBEM framework (Li et al., 2025).

4.3. Output fields

4.3.1. Baseline energy consumption

The baseline energy performance of buildings is evaluated using the *AutoBEM* platform, which generates detailed energy models and simulation outputs for individual buildings. These models are in OpenStudio (.osm) format and EnergyPlus (.idf) format, including comprehensive information such as building geometry, thermal envelope characteristics, internal heat gains, HVAC system configurations, and operational schedules. These models are then simulated using EnergyPlus engine. EnergyPlus performs time-resolved simulations at hourly or sub-hourly intervals, generating end-use energy consumption for heating, cooling, lighting, equipment, and ventilation, as shown in Table 8. These outputs provide the basis for calculating EUI, peak demand, and emissions at both the building and community scales.

4.3.2. Anthropogenic heat emissions assessment

Anthropogenic heat emissions (AHE) represent heat released from buildings to the surrounding outdoor environment through envelope heat exchange, air leakage, mechanical exhaust, and HVAC heat rejection. In *AutoBEM*, AHE is calculated as a post-processing output from EnergyPlus heat-balance and HVAC simulation variables. These estimates quantify building-level heat release to the outdoor environment, but they are not dynamically coupled back to the simulation weather file or to a local microclimate model in the current implementation (Hong, Ferrando et al., 2020; Li et al., 2026).

The simulation outputs from *AutoBEM* provide detailed time-series data on heat transfer variables that contribute to environmental heat loading. Table 9 summarizes the key output variables extracted to calculate AHE from HVAC systems and building components.

Fig. 4 summarizes the AHE post-processing workflow. First, *AutoBEM* runs the EnergyPlus simulation for each building. Second, annual heat-release outputs are extracted for four pathways: exterior surface heat emission, zone exfiltration heat loss, zone exhaust air heat loss, and HVAC heat rejection. Third, these pathway-level outputs are summed to calculate total annual AHE for each building. Finally, annual AHE is normalized by building footprint area or floor area to obtain an AHE intensity metric for comparison across buildings of different sizes.

For building i, annual AHE is calculated as the sum of the annual heat-release components:

$$\mathrm{AHE}_i = E_{\mathrm{surf},i} + E_{\mathrm{exf},i} + E_{\mathrm{exh},i} + E_{\mathrm{rej},i} \quad (1)$$

where $E_{\mathrm{surf},i}$ is annual surface outside-face heat emission, $E_{\mathrm{exf},i}$ is annual zone exfiltration heat loss, $E_{\mathrm{exh},i}$ is annual zone exhaust air heat loss, and $E_{\mathrm{rej},i}$ is annual HVAC heat rejection. All terms are expressed in GJ. The sign convention is applied so that heat released from the building to the outdoor environment is treated as positive.

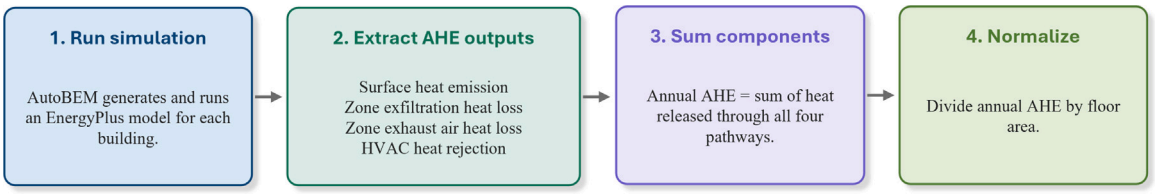


Fig. 4. Anthropogenic heat emission post-processing workflow in *AutoBEM*.

Table 9
AutoBEM output parameters used to estimate annual anthropogenic heat emissions.

AHE pathway	Post-processed output	Description
Envelope heat emission	Surface outside face heat emission [GJ]	Heat released from exterior building surfaces to the surrounding outdoor environment through surface heat exchange.
Zone exfiltration	Zone exfiltration heat loss [GJ]	Heat carried from the building to the outdoor environment by unintentional air leakage through the envelope.
Zone exhaust air	Zone exhaust air heat loss [GJ]	Heat removed from the building by mechanically exhausted air through ventilation or exhaust systems.
HVAC heat rejection	Heat rejection total energy [GJ]	Waste heat discharged by HVAC systems, including condensers, cooling towers, and other heat-rejection equipment.

Anthropogenic heat emission density (AHED) is calculated by normalizing annual AHE by building footprint area:

$$\mathrm{AHED}_i = \frac{\mathrm{AHE}_i}{A_{i,\mathrm{area}}} \tag{2}$$

where $A_{i,\mathrm{area}}$ is the building floor area. AHED supports comparison of heat-release intensity across buildings of different sizes and is used for spatial mapping of building-level heat-emission patterns. When floor-area-normalized comparison is needed, the same annual AHE value can also be divided by total conditioned floor area.

This AHE calculation is a post-processing estimate of heat released from each building to the outdoor environment. It does not modify the ambient weather conditions used by EnergyPlus and does not represent feedback between building waste heat, local air temperature, surface temperature, or urban canopy conditions. Therefore, the current AHE outputs are appropriate for mapping and comparing building-level heat-release patterns, but evaluating feedback effects on local microclimate would require coupling *AutoBEM* with an urban climate or microclimate model.

4.3.3. CO_2 equivalent emission (CO_2-eq) evaluation

The *AutoBEM* framework quantifies building-related greenhouse gas (GHG) emissions by accounting for both natural gas consumption and electricity use (Li et al., 2025). Electricity-related GHG emissions represent upstream emissions from power generation, accounting for the mix of fuels and technologies used in the regional electric grid that supplies the building. Emissions are evaluated as carbon dioxide equivalents (CO_2-eq), incorporating the impacts of carbon dioxide (CO_2), methane (CH_4), and nitrous oxide (N_2O) based on their respective global warming potentials (GWPs). Fig. 5 demonstrates the workflow for Building CO_2-eq Emission Evaluation.

1. Natural Gas Emissions

Emissions from natural gas combustion are calculated using the building's total natural gas consumption and a standardized emission factor:

$$\mathrm{CO2}_{\mathrm{gas}} = \text{Natural Gas (J)} \times \text{Emission Factor} \tag{3}$$

where the natural gas emission factor is 5.613×10^{-8} kg/J (Krey et al., 2014).

2. Electricity-Based Emissions

Electricity-related emissions are estimated using subregion-specific emission factors for CO_2, CH_4, and N_2O, provided by the U.S. EPA's GHG Emission Factors Hub (Environmental Protection Agency (EPA), 2025), which are based on the eGRID subregion of each building's location:

$$\mathrm{CO2}_{\mathrm{elec}} = \mathrm{CO}_2 \text{ factor} \times \text{Electricity (kWh)} \tag{4}$$

$$\mathrm{CH4}_{\mathrm{elec}} = \mathrm{CH}_4 \text{ factor} \times \text{Electricity (kWh)} \tag{5}$$

$$\mathrm{N2O}_{\mathrm{elec}} = \mathrm{N}_2\mathrm{O} \text{ factor} \times \text{Electricity (kWh)} \tag{6}$$

Table 10
100-Year Global Warming Potential values based on EPA (2025) (Environmental Protection Agency (EPA), 2025).

Gas	100-Year Global Warming Potential (GWP)
CH_4	28
N_2O	265

CH_4 emissions account for both incomplete combustion and fugitive emissions during power generation, while N_2O emissions result from high-temperature combustion in fossil fuel and biomass power plants.

These are then converted to CO_2-equivalent using global warming potential (GWP) multipliers (e.g., GWP(CH_4) = 28, GWP(N_2O) = 265, as shown in Table 10):

$$\mathrm{CO2\text{-}eq}_{\mathrm{elec}} = \mathrm{CO2}_{\mathrm{elec}} + \mathrm{CH4}_{\mathrm{elec}} \times \mathrm{GWP}_{\mathrm{CH4}} + \mathrm{N2O}_{\mathrm{elec}} \times \mathrm{GWP}_{\mathrm{N2O}} \tag{7}$$

3. Total CO_2 Equivalent Emissions

Finally, total building emissions are calculated by summing natural gas and electricity-related CO_2-eq contributions:

$$\mathrm{CO2\text{-}eq} = \mathrm{CO2\text{-}eq}_{\mathrm{elec}} + \mathrm{CO2}_{\mathrm{gas}} \tag{8}$$

5. Results and discussion

A case study for buildings in Phoenix, Arizona was carried out using *AutoBEM* to demonstrate the application of the national workflow in a local urban context. Phoenix was selected because its hot-dry climate, high air-conditioning demand, and rapidly growing building stock provide a useful test case for cooling-dominated energy analysis. The results from Phoenix are not intended to represent the full range of U.S. climates, utility prices, grid emissions factors, or building-stock compositions. Rather, the case study demonstrates how the national *AutoBEM* framework can be applied locally to evaluate baseline energy use, emissions, retrofit performance, and future-weather impacts. The same workflow is transferable to other U.S. regions, but the quantitative

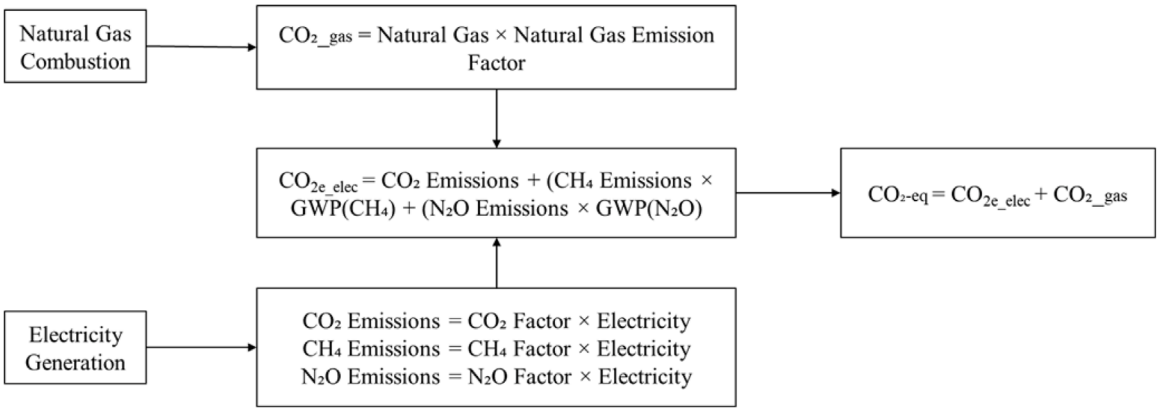


Fig. 5. Workflow for building CO_2-eq Emission Evaluation.

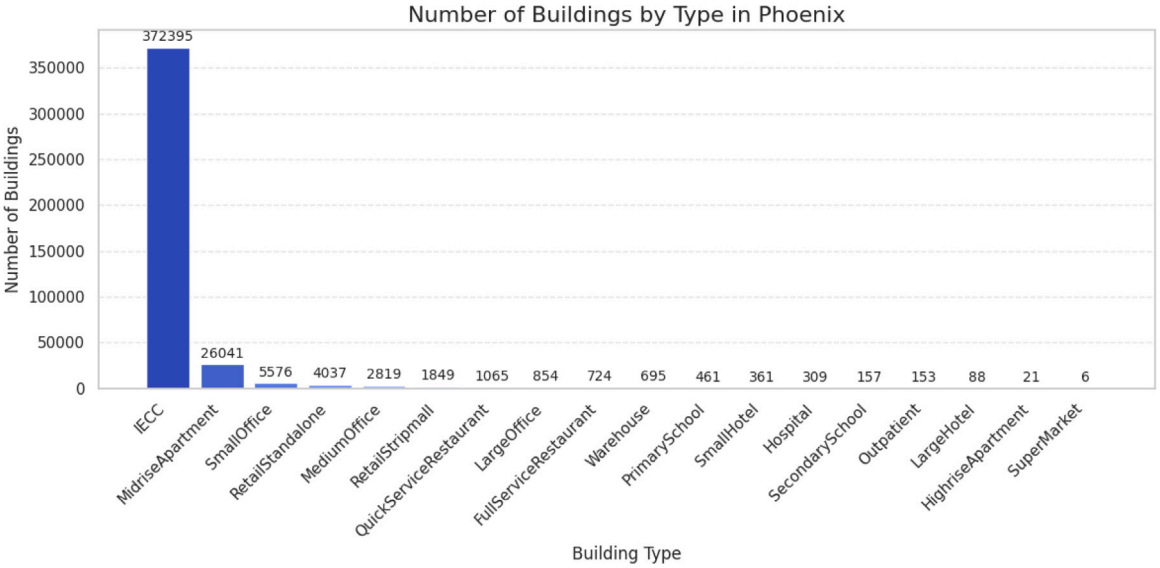


Fig. 6. Number of buildings by type in Phoenix.

results and retrofit rankings are expected to vary by local climate, building characteristics, energy prices, and emissions factors.

Figs. 6 and 7 highlight the distribution of building types by count and total floor area, respectively. Residential buildings dominate the landscape, with over 372,000 IECC homes and 26,000 midrise apartments. However, larger commercial buildings such as large offices, hospitals, and hotels, while relatively few in number, contribute disproportionately to total floor area and, consequently, energy consumption and CO_2 emissions. This underscores the strategic value of targeting large building types for retrofit interventions, as they represent significant opportunities for emission reduction despite their lower prevalence.

Baseline energy consumption, building-stock characterization, and CO_2-equivalent emissions are derived from individualized building-level *AutoBEM* simulations. *AutoBEM* is also capable of performing retrofit simulations at the individual building level. However, because evaluating 151 retrofit measures across millions of buildings would require substantial computational resources, the Phoenix retrofit analysis presented here adopts a representative-archetype scaling strategy as a computational acceleration approach for demonstration.

5.1. Baseline energy consumption

AutoBEM simulations for Phoenix buildings reveal key patterns in baseline energy use across building types. Fig. 8 shows that IECC and MidriseApartment buildings dominate total energy consumption, using roughly 30,000–33,000 TJ of electricity and up to 7000 TJ of natural gas. LargeOffice buildings rank next with high electricity use and minimal gas demand. Commercial types such as restaurants and retail facilities form a secondary tier, each consuming 5000–7000 TJ of electricity with notable gas usage. Other building types, including schools, hospitals, and small offices, contribute modestly. Overall, electricity use far exceeds gas across all categories, though natural gas remains significant in residential and restaurant sectors, underscoring opportunities for electrification and HVAC efficiency improvements.

However, when normalized by building floor area, a different trend emerges (Fig. 9). QuickServiceRestaurant and FullServiceRestaurant buildings show the highest energy use intensities (EUIs) at 5.28 and 4.24 GJ/m^2, respectively, followed by Hospital (2.86 GJ/m^2), Outpatient, and SuperMarket facilities. These energy-intensive building types, though fewer in number, represent prime candidates for targeted efficiency measures. In contrast, IECC residences and Warehouse buildings exhibit the lowest EUIs (0.39 and 0.15 GJ/m^2), indicating relatively efficient performance per unit area despite their large aggregate energy use. This emphasizes that residential and warehouse efficiency gains hinge more on deployment scale than on per-building intensity improvements.

Together, these insights underscore the need for a dual-pronged retrofit strategy: (1) targeting large, high-usage building types (e.g., offices and apartments) for maximum emissions and energy reduction,

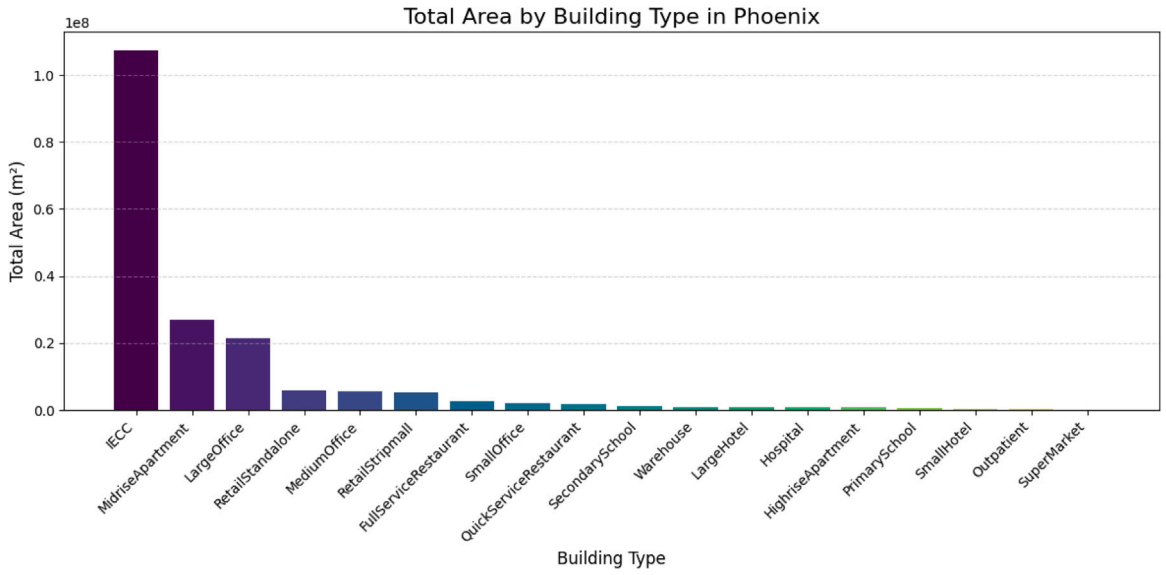


Fig. 7. Total area by building type in Phoenix.

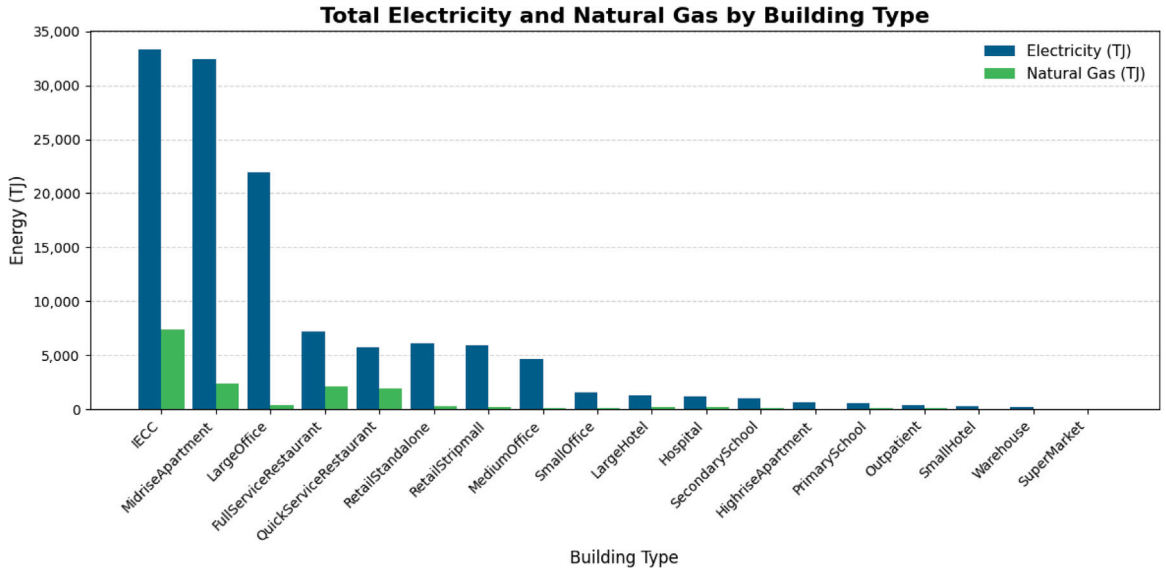


Fig. 8. Total electricity and natural gas by building type in Phoenix.

and (2) addressing high-EUI facilities such as restaurants and hospitals with specialized measures to lower per-unit intensity.

5.2. CO_2 equivalent emission (CO_2-eq) evaluation

In addition to the baseline energy consumption, *AutoBEM* simulations provide detailed quantification of total and intensity-based CO_2-equivalent emissions across the Phoenix building stock. As shown in Fig. 10, the highest total emissions are attributable to IECC residential buildings and midrise apartments, followed closely by large office buildings. These three categories collectively account for the majority of urban emissions, reflecting their combined high energy use and large floor area.

However, analyzing total emissions alone obscures important variation in building-level efficiency. Fig. 11 shows the distribution of CO_2 emission intensity by building type. Quick-service restaurants (exceeding 40 kg CO_2/ft^2), full-service restaurants, and hospitals exhibit the highest emission intensities. These facility types are consistently high emitters regardless of size, highlighting the need for targeted efficiency measures tailored to their unique usage patterns and equipment loads.

Further stratification by construction vintage reveals the impact of building standards on emissions performance. Fig. 12 presents a heatmap of average emission intensity by building type and ASHRAE standard. Older vintages (e.g., DOE-Ref-Pre-1980 and 1980–2004) show significantly higher emissions across nearly all building types, especially for restaurants, hospitals, and retail standalone buildings. In contrast, buildings constructed under more recent standards (e.g., ASHRAE 90.1–2013 and later) generally perform better, though gaps remain in several commercial categories.

Together, these results demonstrate the value of emissions disaggregation: while high-volume residential buildings dominate in absolute terms, commercial buildings, especially those with outdated HVAC and envelope systems, represent high-priority targets for decarbonization. The *AutoBEM* framework enables this type of nuanced evaluation to inform retrofit prioritization and policy planning.

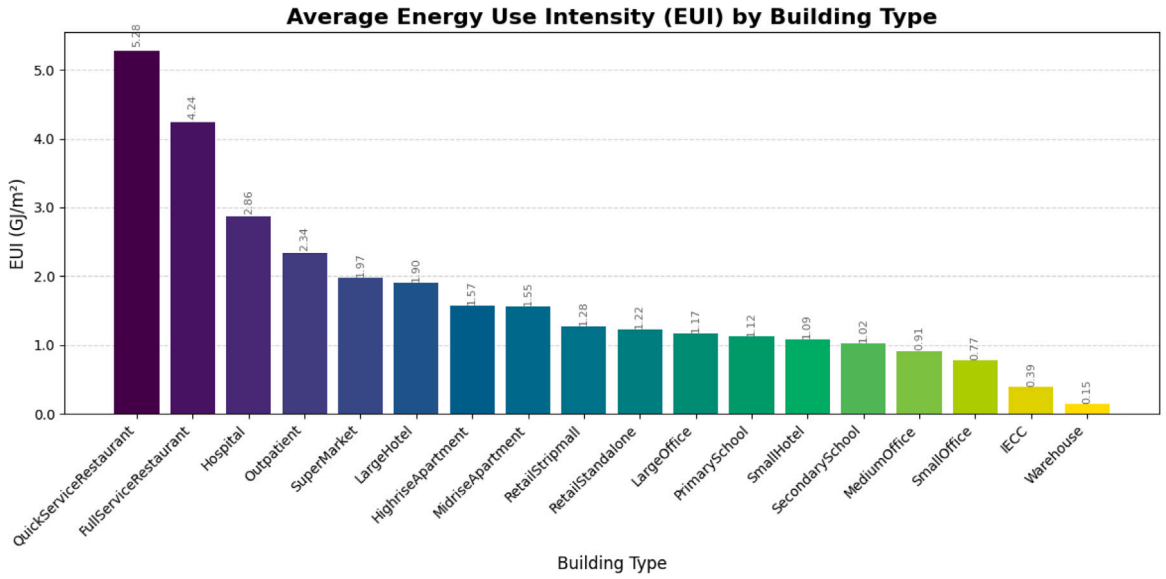


Fig. 9. EUI by building type in Phoenix.

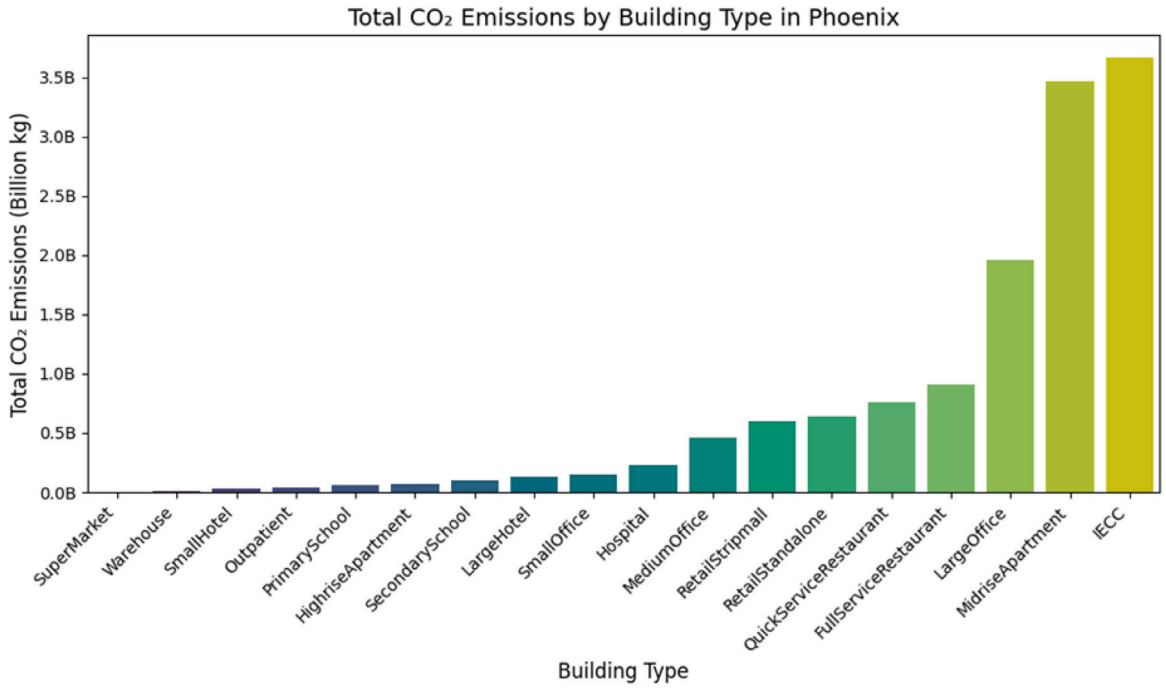


Fig. 10. Total CO2 emissions by building type in Phoenix.

5.3. ECM evaluation

For each building type and construction standard group, a representative archetype based on medium area was selected (Chowdhury, Li, Stubbings, New, Garg et al., 2023; Garg et al., 2024). 151 retrofit measures were then applied across 133 representative building archetypes in Phoenix. Each archetype serves as a proxy for a broader segment of the building stock. Area multipliers were used for each archetype to reflect the whole city energy savings and cost savings. Therefore, retrofit results in this case study are archetype-scaled estimates designed for computationally efficient city-scale screening, rather than direct retrofit simulations for every individual building. Scalable simulation and comparative analysis of retrofit performance were then conducted across different usage types and energy codes.

Table 11 shows an example measure for window upgrade. The retrofit technology details and cost information were collected from National Residential Efficiency Measures Database (National Laboratory of the Rockies (NRL), 2025). The window upgrade measure was then applied to the archetype buildings in Phoenix for simulation in *AutoBEM*.

By comparing the simulation outputs to the baseline, *AutoBEM* generated estimated paybacks and CO_2 reductions. Fig. 13 presents the top five building types and construction standards in Phoenix with the shortest payback periods for the window upgrade measure targeting a U-Value of 0.482. The shortest payback occurs for *MediumOffice, 90.1-2007* at 1.45 years, followed by *Hospital, 90.1-2010* at 1.50 years. IECC residential archetypes built to *90.1-2007* and *90.1-2013* achieve 1.55 and 1.59 years, respectively, while *Hospital, 90.1-2016* is similarly favorable at 1.59 years. Overall, it indicates that the window upgrade

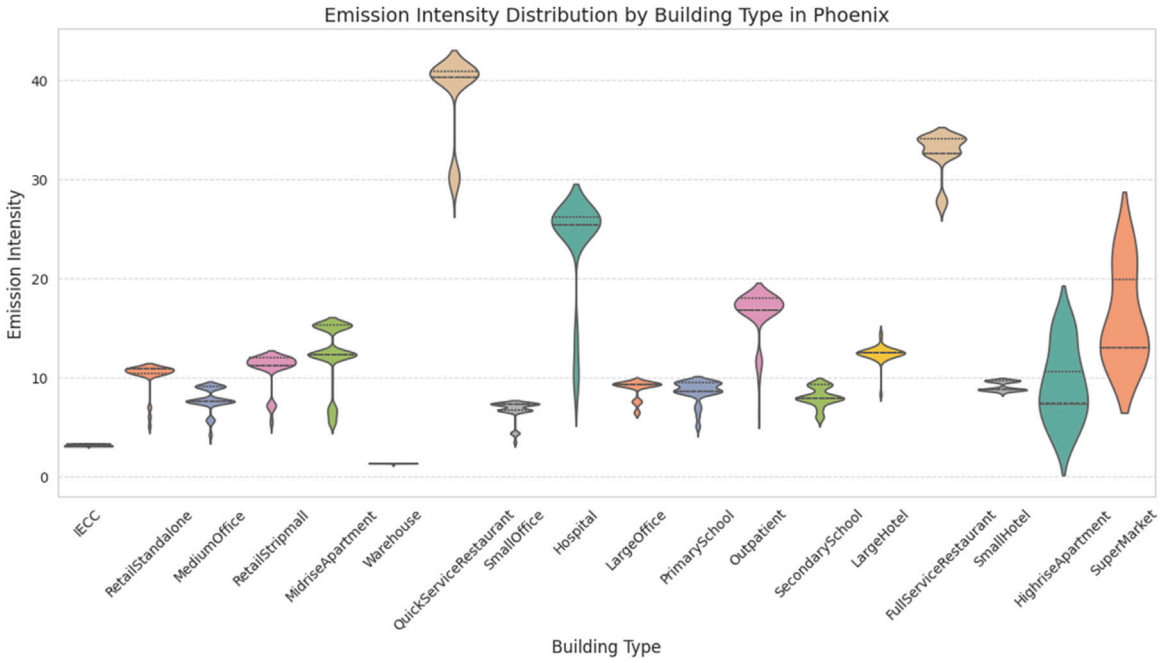


Fig. 11. Emission intensity distribution by building type in Phoenix (kg CO_2/ft^2).

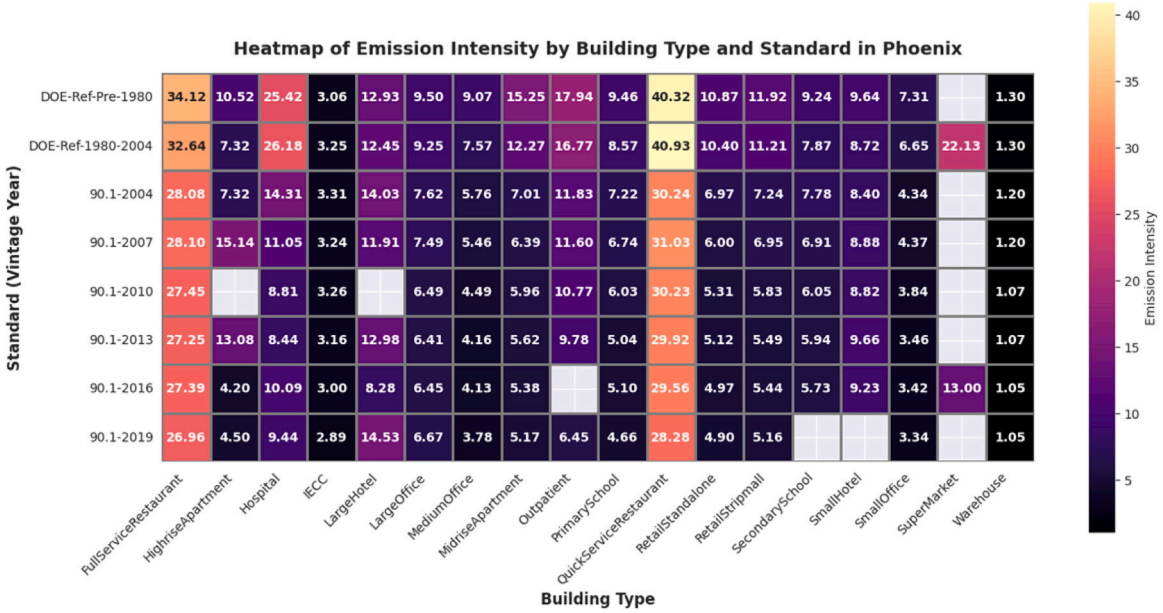


Fig. 12. Heatmap of emission intensity by building type and standard in Phoenix (kg CO_2/ft^2).

Table 11
Measure details for window upgrade (National Laboratory of the Rockies (NRL), 2025).

Measure name	Retrofit technology details	Unit cost
Upgrade Window U-Value to 0.482	Single-Pane, Clear, Non-metal Frame, with Clear Storm • Frame Material: non-metal • Glazing Type: clear • Panes: 1 • SHGC: 0.555 • Storm Sash: clear • U-Value: 0.482 Btu/h-ft^2-° F (2.74 W/m^2 K)	\$118.4/m^2

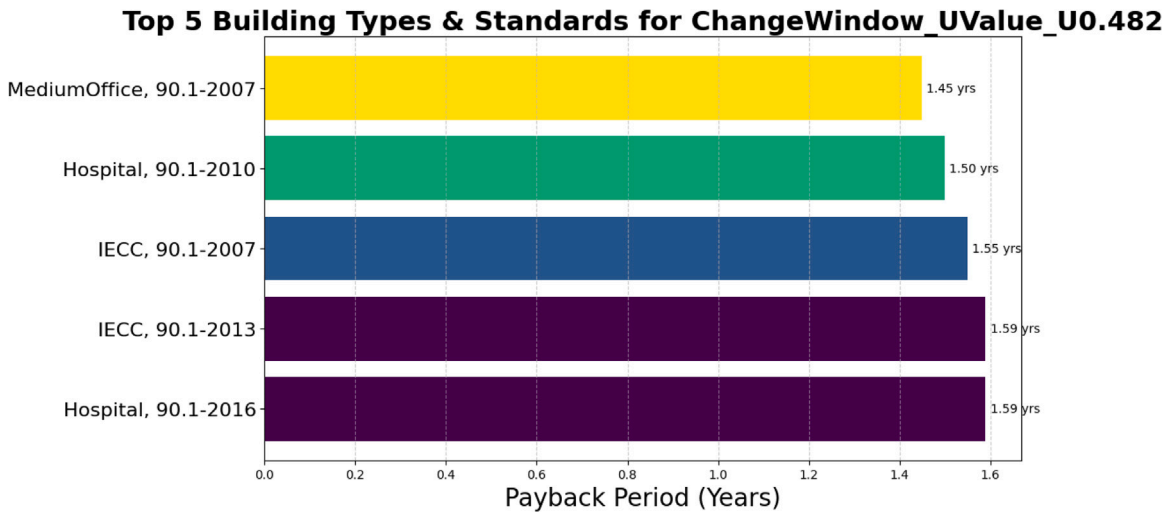


Fig. 13. Top 5 building types and standards for window upgrade (U-Value = 0.482) by payback.

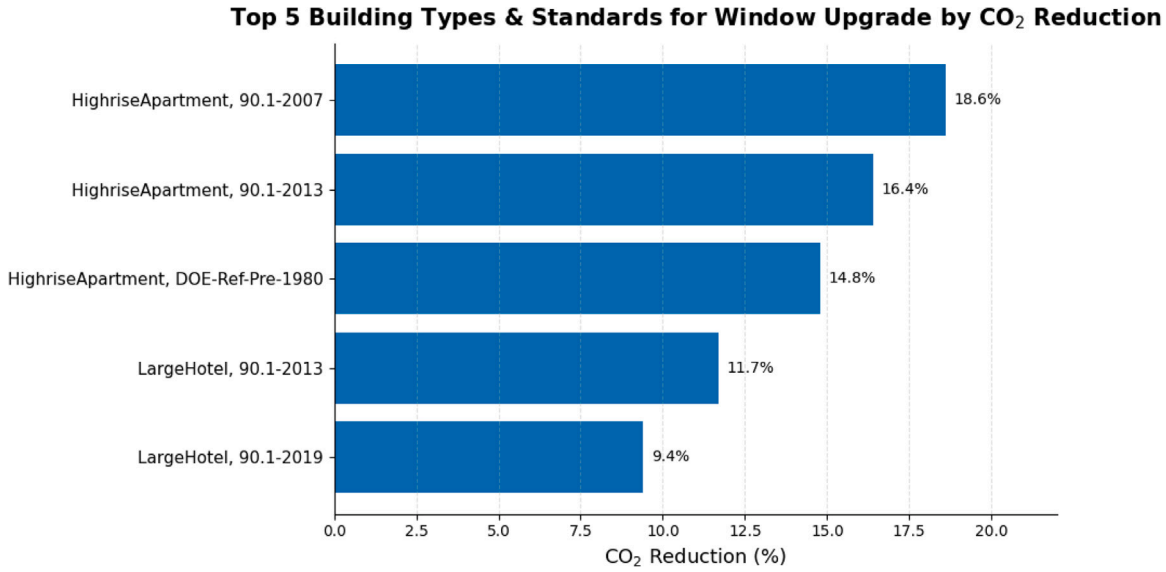


Fig. 14. Top 5 building types and standards for window upgrade (U-Value = 0.482) by CO_2 reduction.

yields relatively rapid returns across office, residential (IECC), and hospital archetypes in Phoenix's hot-dry climate.

Fig. 14 highlights the top five building types and standards achieving the highest percentage reduction in CO_2 emissions from implementing the window upgrade (U-Value = 0.482). The HighriseApartment, 90.1-2007 led with an average CO_2 reduction of 18.6%, closely followed by HighriseApartment, 90.1-2013 at 16.4% and HighriseApartment, DOE-Ref-Pre-1980 at 14.8%. Two LargeHotel vintages also ranked in the top five. These results underscore the significant emissions mitigation potential of improved window performance, especially in multifamily and large commercial buildings in Phoenix's hot-dry climate.

While the archetype-level analysis provides detailed insight into the performance of retrofit measures for representative buildings, scaling these results to the city level requires accounting for the prevalence and footprint of each archetype. To enable this, area multipliers were applied to each archetype based on its total building stock representation in Phoenix. These multipliers were used to compute aggregate investment costs, energy savings, and emissions reductions for each retrofit measure across the full city. The resulting city-level dataset captures the scaled impact of all 151 retrofit measures across 133 archetypes.

By comparing retrofit measures across all simulated building archetypes in Phoenix, *AutoBEM* identified the best-performing building type and standard for each measure in terms of payback period. Fig. 15 presents the top 20 retrofit measures ranked by the shortest payback periods. The leading measures are dominated by HVAC system upgrades, particularly configurations involving packaged terminal air conditioners (PTAC), unit heaters, and evaporative cooling systems, which achieve paybacks around 1.5 years. Hospitals built to the DOE-Ref-1980–2004 standard consistently appear among the top performers across multiple retrofit categories, reflecting their high baseline energy intensity and strong savings potential. Other efficient combinations include high-rise apartments (90.1–2007) and large hotels (90.1–2019), both benefiting from improved HVAC configurations such as dedicated outdoor air systems (DOAS) with heat recovery or variable refrigerant flow systems.

Under the standardized cost and utility-rate assumptions used in this screening analysis, the most economically favorable retrofit measures in Phoenix are primarily high-efficiency HVAC upgrades, including packaged terminal systems, evaporative coolers, and unit heaters. Their short modeled payback periods reflect the strong influence of cooling loads and equipment efficiency in Phoenix's hot-dry climate. Envelope

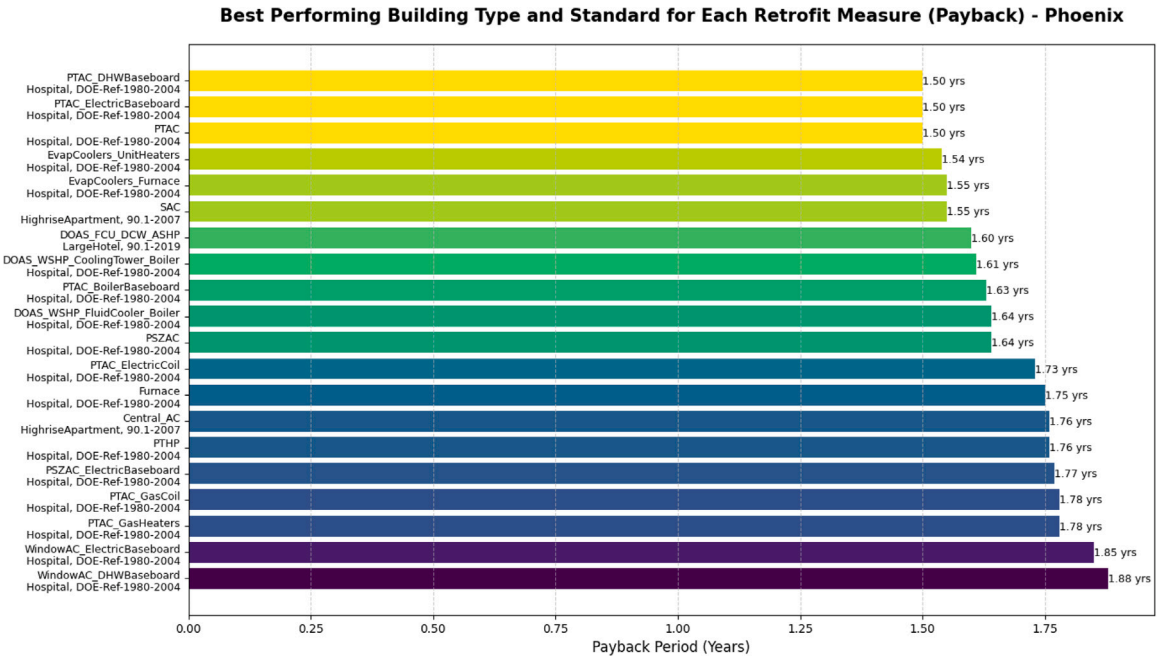


Fig. 15. Best-performing building type and standard for each retrofit measure ranked by payback period in Phoenix (ranked by payback).

and insulation measures generally show less immediate payback potential, but can still provide long-term energy and emissions benefits, particularly when combined with equipment upgrades.

The short payback periods identified for the top-performing retrofit cases should also be interpreted in light of prior empirical evidence. For example, Liang et al. analyzed realized post-retrofit electricity savings from the Energize Phoenix program and found that actual savings were substantially lower than engineering-model predictions, with realized savings approximately 30%–50% below predicted values (Liang et al., 2018). They reported average simple payback periods of about 2.7 years for commercial buildings and much longer paybacks for residential buildings. In comparison, the results presented here are engineering-model-based estimates for representative archetypes under idealized implementation assumptions and localized utility rates. Therefore, the short paybacks shown in Fig. 15 are modeled technical-economic screening results rather than actual measured post-retrofit performance. In hot-dry climates such as Phoenix, certain HVAC-focused measures can produce large modeled savings for specific high-intensity archetypes, especially when evaluated using prototype-consistent schedules, autosized systems, and nominal performance assumptions. The short payback periods identified for some Phoenix archetypes represent modeled technical-economic screening results under standardized assumptions. They should be used to prioritize promising building-measure combinations for further investigation, rather than as final estimates of realized project payback for individual buildings. Real-world performance may be lower due to project-specific installation complexity, behavioral effects, installation quality, equipment operation, soft costs, maintenance, incentives and other implementation factors not captured in the present screening framework.

The best-performing measures are highly dependent on local climate conditions, building characteristics, and utility pricing structures. For example, in a colder climate, heating-focused measures such as boiler upgrades or wall insulation with higher R-values might yield better returns. Similarly, variations in electricity and gas rates can significantly alter payback periods and net savings. Measures that are economically favorable under one utility rate structure may be less effective under another. Therefore, while the Phoenix results offer valuable insights, effective retrofit prioritization must be region-specific, accounting for local climate, building stock characteristics, and energy economics. *AutoBEM* enables precisely this kind of tailored analysis, supporting data-driven, location-sensitive decision-making for scalable and impactful energy retrofits.

5.4. Future weather exploration

To demonstrate *AutoBEM*'s capability to simulate future energy demand under changing climatic conditions, fTMY weather files were integrated into the modeling workflow for several cities in the United States. These fTMY datasets generated for four SSP–RCP scenarios enable projection of building-level energy use and emissions across long-term climate horizons (e.g.,2020–2040, 2040–2060, 2060–2080, and 2080–2100).

As a representative example, Fig. 16 shows the *AutoBEM* web interface used to explore fTMY-based simulation results for Phoenix. Users can interactively select individual buildings and view projected changes in annual energy use, cost, and CO_2-equivalent emissions. Percent change values are calculated relative to the 2020–2040 baseline, and results are disaggregated into electricity and natural gas components.

The sample building in the visualization exhibits a steady increase in electricity demand over time — up to 11.3% higher by 2080–2100 — reflecting rising cooling needs in a warming climate. Conversely, natural gas consumption shows a decline of over 32%, driven by reduced heating demand. These results underscore the potential for significant shifts in energy portfolios under climate change and emphasize the need for adaptive strategies in energy planning.

5.5. Key findings and insights

The Phoenix case study is intended to demonstrate how *AutoBEM* outputs can support screening- and planning-level decisions, rather than to derive universal retrofit conclusions. The value of the results lies in the ability to combine building-resolved baseline simulations, ECM analysis, emissions calculations, local utility prices, and future-weather scenarios into a consistent decision-support workflow. Three application-oriented findings emerge from the Phoenix demonstration.

First, the results show that different decision metrics identify different priority building groups. Total energy and emissions results



Fig. 16. *AutoBEM* web-based visualization of fTMY-based future energy projections for Phoenix, AZ. The interface shows building-level projections for energy consumption, cost, and emissions across multiple time horizons using SSP5–RCP8.5 scenario.

identify building types that dominate citywide demand because of their large stock size or floor area, including residential buildings, MidriseApartment buildings, and LargeOffice buildings. These categories are appropriate targets for broad portfolio-scale policies, such as mass-market incentive programs, utility demand-side management planning, or neighborhood-scale decarbonization strategies. In contrast, intensity-based metrics identify smaller but more energy-intensive building types. QuickServiceRestaurant and FullServiceRestaurant buildings have the highest EUIs, at approximately 5.28 and 4.24 GJ/m^2, respectively, followed by Hospital buildings. Similarly, emissions-intensity results identify restaurants and hospitals as high-priority commercial targets. This distinction is important because a program designed only around total emissions would prioritize large stock segments, while a program designed around energy or emissions intensity would prioritize more specialized commercial uses.

Second, the ECM results provide a measure-specific screening framework rather than a single generic retrofit recommendation. For the window upgrade example, the shortest paybacks occur for specific building type and standard combinations, including MediumOffice under 90.1–2007 and Hospital under 90.1–2010. In contrast, the largest CO_2 reductions for the same window measure occur for HighriseApartment and LargeHotel archetypes. The broader ranking of retrofit measures shows that many of the shortest modeled paybacks in Phoenix are associated with HVAC-related measures, particularly for high-intensity commercial archetypes. These results can be used to identify candidate building-measure combinations for further investigation, such as targeting audits, designing incentive eligibility criteria, prioritizing utility program outreach, or selecting measures for more detailed calibrated modeling.

Third, the fTMY results demonstrate how the same workflow can be used for climate-adaptive planning. Under the future-weather scenario illustrated for Phoenix, annual electricity demand increases while natural gas demand decreases, reflecting increased cooling demand and reduced heating demand under a warmer climate. This type of result can help planners evaluate whether measures that perform well under TMY conditions remain valuable under future climate conditions. It also helps identify potential shifts in utility load, fuel use, emissions, and retrofit economics over long planning horizons.

These outputs should be interpreted as screening-level and planning-level results. They are useful for identifying high-impact building groups, prioritizing candidate measures, comparing policy or utility-program scenarios, and selecting buildings or archetypes for more detailed follow-up analysis. They are not intended to replace building-specific audits, calibrated utility bill analysis, engineering design, or investment-grade retrofit evaluation. Project-level decisions require additional building-specific data, measured energy use, local cost estimates, engineering review, and consideration of installation constraints, incentives, maintenance, and occupant behavior. Therefore, the Phoenix case study demonstrates the type of decision-support workflow enabled by *AutoBEM*, while the specific retrofit rankings, paybacks, and emissions reductions should be considered within the local climate, building-stock, utility-price, and modeling assumptions used in this analysis.

Overall, these results demonstrate the value of combining building-level simulation with scalable analysis to support more targeted and climate-informed retrofit decision-making. The generalizability of these results lies primarily in the workflow rather than in the Phoenix-specific numerical outcomes. *AutoBEM* can be applied to other regions using local building-stock data, weather files, utility prices, and grid-emissions factors. However, retrofit rankings, payback periods, emissions reductions, and future-weather impacts are expected to differ across climate zones and building-stock compositions. For example, HVAC cooling measures are likely to be more favorable in hot-dry climates such as Phoenix, whereas envelope insulation, heating-system upgrades, and electrification strategies may become more important in colder or mixed climates. Therefore, the Phoenix case study is a demonstration of the framework's capability rather than a basis for universal national retrofit conclusions.

5.6. Limitations and interpretation of results

The national-scale scope of *AutoBEM* requires several simplifying assumptions that introduce uncertainty into the simulation results. These assumptions include single-zone thermal representation, area-weighted internal load and space-type parameters, inferred building type, estimated building height, estimated construction year, and prototype-derived schedules, envelope properties, and HVAC configurations. These simplifications are necessary for generating and simulating models across millions of buildings, but they affect the interpretation of both baseline and retrofit results.

First, representing each building as a single thermal zone can smooth spatial differences in internal loads, solar exposure, occupancy, and HVAC operation. This may reduce the model's ability to capture simultaneous heating and cooling, perimeter-core load differences, and

localized peak demand. As a result, single-zone models are better suited for annual or aggregate energy estimation than for detailed zone-level comfort, equipment sizing, or peak-load analysis. The use of area-weighted parameters further supports scalability, but it may obscure heterogeneity among space types within mixed-use or complex buildings. Ongoing work within the *AutoBEM* is addressing this limitation through an *AutoZone* workflow that automatically subdivides single-zone building geometries into multi-zone representations. Multi-zone models can better represent perimeter-core load differences, but they require substantially greater computational resources. Therefore, the present study adopts single-zone models for national-scale computational feasibility, while future work will evaluate targeted use of automated zoning where additional fidelity is most beneficial.

Second, uncertainty in inferred building type can affect multiple downstream assumptions, including occupancy schedules, internal gains, HVAC system selection, ventilation rates, envelope characteristics, and retrofit applicability. For example, misclassifying a restaurant, retail building, or office may lead to substantial differences in simulated energy use intensity because these building types have different equipment loads, operating schedules, and ventilation requirements. Similarly, uncertainty in estimated construction year affects assignment of code vintage, envelope performance, and system characteristics, which can influence both baseline energy consumption and estimated retrofit savings.

Third, building height uncertainty affects estimated number of floors and total floor area. Because many energy, emissions, and cost results scale with floor area, height errors can directly affect aggregate energy consumption, CO_2-equivalent emissions, anthropogenic heat emissions, and retrofit investment cost. These effects are particularly important for multi-story commercial and multifamily buildings. In contrast, normalized metrics such as EUI or emission intensity may be less sensitive to total floor-area errors but remain sensitive to building type, vintage, schedules, and system assumptions.

These uncertainty sources help explain why *AutoBEM* results should be interpreted differently across scales. As discussed in Section 1.1, individual-building outputs from uncalibrated models can exhibit substantial error because many building-specific characteristics are not directly observed. However, confidence generally improves when results are aggregated by neighborhood, city, building type, construction vintage, or utility territory, because individual-building errors can partially offset across large samples. Therefore, the results presented in this case study are intended for screening- and planning-level interpretation rather than calibrated prediction for individual buildings. At the stock or portfolio scale, *AutoBEM* outputs can help cities, utilities, policymakers, and energy-program administrators identify high-impact building segments, compare relative retrofit opportunities, prioritize candidate measures, and evaluate how energy, emissions, and payback outcomes vary with building type, construction standard, climate, utility prices, and grid emissions factors. For example, total energy and emissions results can highlight building categories that dominate aggregate demand, while EUI and emission-intensity metrics can identify smaller but energy-intensive building types that may warrant targeted programs. Archetype-scaled ECM results should therefore be used to prioritize promising building-measure combinations for further investigation, not as final investment-grade estimates for individual buildings. Project-level decisions require additional building-specific data, measured energy use, engineering review, local cost estimates, and, where possible, calibrated models.

A full probabilistic uncertainty propagation across all input assumptions, building models, climate scenarios, and 151 ECMs is beyond the scope of the present study. Nevertheless, the validation evidence and the sensitivity of model outputs to building type, height, vintage, and zoning assumptions indicate that uncertainty should be explicitly considered when interpreting the results. Future work will incorporate formal sensitivity and uncertainty analyses by perturbing key input variables, including building type, height, vintage, conditioned floor area, zoning representation, utility prices, and ECM cost and performance assumptions. Such analysis will help quantify confidence intervals for baseline energy use, emissions, retrofit savings, and payback estimates across different building categories and spatial scales.

6. Conclusion

This paper presents *AutoBEM*, a nationwide framework for automated, bottom-up building energy modeling and retrofit evaluation across the United States. By leveraging the Model America dataset, high-performance computing, and EnergyPlus engine, *AutoBEM* enables the generation and simulation of building energy models for over 122 million structures, representing 97.8% of the national building stock.

AutoBEM supports scalable assessments of energy consumption, CO_2 and anthropogenic heat emissions, and retrofit performance, incorporating typical and future climate scenarios through TMY and fTMY weather data. By simulating and evaluating 151 ECMs with localized utility pricing, *AutoBEM* offers an automated and reproducible framework to quantify energy savings, cost-effectiveness, and environmental benefits at the individual building, district, or national scale.

A case study in Phoenix, Arizona demonstrated how *AutoBEM* can be used to identify locally relevant retrofit strategies under hot-dry climate conditions and Phoenix-specific building-stock and utility assumptions. The results underscored the dual importance of targeting both high-energy-use building types and high-EUI facilities, and highlighted the sensitivity of retrofit performance to climate and utility pricing variability—challenges that *AutoBEM* is designed to address. Future work will apply the same workflow across additional climate zones, utility territories, and building-stock compositions to quantify regional variability in retrofit performance and climate sensitivity.

Methodologically, *AutoBEM* integrates building-resolved nationwide UBEM, scalable ECM analysis, and fTMY weather scenarios within one reproducible pipeline. Practically, this provides policy-relevant decision-support outputs, including ECM payback, cost effectiveness, and CO_2 reduction estimates for screening and prioritization across U.S. communities.

As a step toward a national digital twin for the built environment, *AutoBEM* improves the spatial resolution, scalability, and accessibility of physics-based building energy modeling across the U.S. building stock. Future development will include expanded datasets (e.g., solar potential, EV readiness, socioeconomic overlays), deeper empirical calibration, and continued refinement of archetype generation and ECM logic. By enabling equitable, cost-effective pathways toward decarbonization, *AutoBEM* helps accelerate the transition to a more energy-efficient, resilient, and sustainable built environment.

CRediT authorship contribution statement

Hang Li: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Shovan Chowdhury:** Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Fengqi Li:** Writing – original draft, Supervision, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Vivek Kumar Singh:** Writing – original draft, Conceptualization. **Brett Bass:** Writing – original draft, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Joshua New:** Writing – original draft, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT (GPT-5) in order to correct grammar errors and improve clarity and readability. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Selected building-level input data and simulation outputs from the Model America v1.0 dataset are publicly available through the public repository described in Bass, New, Berres, Clinton et al. (2022). These public data include national-scale building-stock data to support *AutoBEM* simulation.

Some datasets used in this study are not publicly available. Utility meter data and other validation datasets are subject to data-use agreements, privacy restrictions, or third-party ownership and therefore cannot be redistributed by the authors. Project-specific processed datasets for the Phoenix case study, including retrofit evaluation outputs, are available from the corresponding author upon reasonable request, subject to institutional approval and applicable data-use restrictions.

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